IMPACT OF FIIS' INVESTMENTS IN IPO ON THE INDIAN STOCK MARKET

ABSTRACT

Over the past few decades the global economy underwent through various reforms and deregulations one of which was liberalization of financial markets popularly termed as capital account liberalization. There has been growing debate in the literature regarding capital account liberalization which is allowing foreign investors to participate in the capital markets of one country and vice versa. Though Foreign Direct Investments (FDI) have a long run objective and is often associated with transfer of knowhow, resources etc the Foreign Portfolio flows (FPI) having short run profit maximizing objective is often said to destabilize the economy by its volatile buying and selling behavior .Post liberalization the Indian economy also went through far reaching changes. One such change was the liberal entry to foreign portfolio investments which are direct investments in the stock market have drawn considerable attention .An attempt has been made in our research to understand the significance of foreign institutional investors(FIIs) which are portfolio flows in the Indian context. The capital markets are broadly classified into secondary market and primary market. While the primary market raises the initial capital of the corporate through mechanisms like IPO (initial public offering), FPO, Private Placements etc, once the capital is raised the shares can then be traded in the secondary market popularly known as the stock market. Our objective was to understand the investments of the FIIs in these IPOs from various perspectives. Evidence suggests that the FII investments in IPO had a significant impact on the market capitalisation and nifty .In addition to that scripts having higher FII investments had the chances of higher short term listing gains.

Key words : Foreign Institutional Investors, Initial public offering, market capitalisation ,nifty, short term profits.

INTRODUCTION

1.1. THE RESEARCH CONTEXT

The last few decades have witnessed the financial liberalization of equity markets across the world. Equity market liberalizations give foreign investors the opportunity to invest in domestic equity securities and domestic investors the right to transact in foreign equity securities. Though studies have found that capital account liberalization played an important role in future economic growth it does not justify the contribution of equity market equally (Bekaert et .all,2005);(Hubbard,1997);(Gilchrist & Himmelberg,1999).Since capital market liberalization at most times necessitates better corporate governance and investor protection it thereby promotes financial development (La Porta et all 1997) and hence growth(King & Levine,1993).But such foreign capital may not always be profitably invested if the underlying foundations of the economy receiving such funds are weak(Stglitz,2000).Thus the growth effects of such flows find mixed results(Eichengreen, 2002).Moreover though such flows have witnessed stock market development but at most times has not resulted into speedier industrialization and economic growth for most emerging market economies(Singh& Wiesse,1998);(Stglitz,2003).It has also been observed through studies that though capital market liberalization does lead to greater financial depth and economic growth, it is mostly so incase of developed economies and less in developing economies (Klein& Olivei,2008).On the contrary studies conducted on emerging market economies have also witnessed huge private investments leading to economic growth.(Henry, 1999).Thus there has been a debatable literature on the positive contributions of capital market liberalisation. Moreover the foreign portfolio flows constituting a major part of such liberalized flows has its own problems because of its links with other macroeconomic parameters and the. nature of such flows. The most outstanding feature of portfolio investment, as opposed to other forms of capital inflows such as direct foreign investment, borrowing from international financial institutions, or long-term bank loans is that they present a potential risk of reversal of flows in a very short term, that is, the possibility that foreign investors may suddenly decide to leave the country they are investing in. This potential risk of flow reversal may be harmful in terms of either a greater exchange rate volatility, greater interest rate volatility or both. Furthermore, if the central bank lacks the necessary speed of reaction and stock of international reserves is at a low level, it may cause a balance of payments crisis.(Agarwal,1997).Thus the Foreign Institutional Investors (FIIs) constituting the bulk amount of such portfolio flows in recent times deserve special attention in terms of contribution in the Indian context. The purpose of the flow of capital to underdeveloped or developing countries is to help in their economic development .Capital flows in the form of private investment, foreign investment; foreign aid and private bank lending are the principle ways by which resources can come from rich to poor countries. The transmission of technology, ideas and knowledge are other special types of resource transfer.These foreign investments can be categorised into various categories like foreign direct investment (FDI), foreign institutional investment (FII), non-resident Indian (NRI) and person of Indian origin (PIO) investment. India opened up to foreign investments gradually over the past two decades, especially since the landmark economic liberalization of 1991. In India foreign institutional investors (FIIs) have been allowed to invest in the domestic financial market since 1992. This decision to open up the Indian financial market to FII portfolio flows at that point in time was influenced by several factors such as the complete disarray in India's external finances in 1991 and a disorder in the country's capital market. International portfolio flows are, as opposed to foreign direct investment, liquid in nature and are motivated by international portfolio diversification benefits for individual and institutional investors in industrial countries. They are usually undertaken by institutional investors like pension funds and mutual funds. Such flows are, therefore, largely determined by the performance of the stock markets of the host countries relative to world markets.

The FII inflows into Indian equity market can be classified in to two ways viz., Primary market and secondary market. The FII inflow to primary market in India comes mainly through the conversion of foreign currency convertible bonds (FCCBs), private placement to qualified institutions placements (QIPs), initial public offers (IPOs), follow-on overseas offers, conversion of warrants and preferential offers.This research is intended to focus upon few significant dimensions and contributions of FII investments in India through the Primary Market.

1.2 LITERATURE REVIEW

Asha C. Parsuna (2000), finds that mainly the return in the host country stock market attract the FIIs investments, other factors are also creating impact on the arrival of FIIs but they are statistically insignificant. *Chakrabarti (2001,)* came with the evidence that the FIIs flows are highly correlated with equity returns in India. He also found that the FII flows effect rather than cause of these returns and hence it contradicted the view that FIIs determine the market return in general. *Mukherjee and*

Coondoo (2002), also explore the relationship of FII Investments to the Indian equity market with its possible covariates based on a daily data set for the period January 1999 to May 2002. The study finds the FIIs flows to and from the Indian market tend to be caused by return in domestic equity market. The study also explains that the return from the exchange rate variation and fundamentals of the Indian economy may have influenced FIIs decision, but such influence does not seem to be strong.

Gorden and Gupta (2003), apply the multiple regression technique on the monthly data from September 1992 to October 2001 to find out the relation between fundamental factors of the Indian economy and portfolio flows and find external interest rate and lagged domestic stock market return as key variables for explaining portfolio arrivals.

Rai & Bhanumurthy (2003), examine the determinants of the FIIs investments in India by taking the data from January 1994 to November 2002 and find a positive relation between FIIs and stock market return (BSE) and an adverse effect of fundamental factors such as speculation and sentiments.

Lakshmi Sharma (2005), framed a model by taking FIIs investment as dependent variable and impact cost, market return and the ratio of non-promoters category of shareholders to total outstanding shares as independent variables and found that impact cost and the quantum of the shares available for trading in the market seem to be two important considerations for FIIs for their investment purpose. But of the two significant variables, impact cost had emerged as the most important variable explaining the FIIs investment in a company.

Panda (2005), tried to examine the impact of FIIs Investments on the Indian stock market by applying VAR analysis on the daily data from October 2003 to March 2004 and found Mutual Fund investments having better explanatory power than FIIs investments in explaining returns on both of the main Indian markets BSE and NSE Nifty. The investigation found that FII investments did not affect BSE Sensex rather it was affected by the later.

Bandhani (2005), studies interlinkage of stock prices, net FIIs investment and exchange rate. Using monthly data from April 1993 to March 2004. He observed bi-directional long-term causality between FIIs investment flows and stock prices, but no short-term causality could be traced between the variables. The author further found that exchange rate long-term granger causes FIIs investment flows, not vice versa. This study does not find any long run relationship between exchange rate and stock prices but short-term causality runs from changes in exchange rate to stock returns, not vice-versa.

Tripathy (2007), examines the interlinkage among stock market, market capitalization and net FII investments by applying both Ganger Causality and Vector Auto Regression test (VAR). The results indicate that there is no significantly causality between FII investment and market capitalization but there is an unidirectional casual relationship between market capitalization and stock market and net FII investment in 38 stock market.

1.3 RESEARCH GAP

In almost all the researches focus has always been laid on FII investment behaviours in the secondary market. Much of the researches were attempted to analyse the various determinants of FII inflows into the Indian secondary market and the impact of FII inflows into secondary market on various macroeconomic parameters. However there has been not much evidence of substantial research on FII investments in the Indian Primary market and its impacts. As we know that the FIIs are major institutions who generally invest in the foreign country for reaping profits for their investors, they are presumed to invest through well researched strategies whether short term or long term. This research is intended in understanding mainly the various ways in which the FII investments in India affect the stock markets. As discussed through review of literature in the previous sections we observe that there has not been much research that we could witness on FII investments in the Indian Primary market. We have mainly focused on FII investments in Indian IPOs and how this is affecting other significant stock market variables.

1.4 RESEARCH OBJECTIVES

In order to probe deeper into the research question the following objectives have been framed:

1. The research will try to study the growth and extent of FII activities in the Indian Financial market.
2. In order to understand the impact of FII investments into Indian IPOs we will first try to analyse the impact of IPO itself on the financial market.
3. We shall further try to analyse the impact of FII inflows into IPO on the Indian financial market.
4. The research will also attempt to understand the impact of FII investment proportions in each stock on the short run listing performance of the respective stocks.

1.5 JUSTIFICATION OF VARIABLES

In order to understand the impact of IPO on financial market, we have considered two important parameters i.e. market capitalization& NIFTY. Market capitalization is computed by multiplying the stock market price per share by the number of shares that are outstanding (that is by the no. of shares not held by the company itself). In terms of economic significance, marker capitalization or a proxy for market size is positively related to the ability to mobilise capital and diversity risk. Similarly NIFTY is a well-diversifie stock index accounting for 23 sectors of the economy and is assumed to be a stormy indicator of market performance. We will be considering both these variables as dependent variables in our research to be representing the financial markets and try to study the impact of FII inflows into IPO on these variables. While doing so we shall be controlling for GDP which is one of the most important economic indicators and in the long run accounts for stock market dvelopement. Thus inspite of controlling for GDP we would try to study the impact of FII inflows into IPOs on financial markets to put forth our objective. Moreover the short run listing gain or loss has also been considered as a dependent variable with FII investment proportion in each stock as independent variable while controlling for market subscription in order to understand by how much it is actually contributing to accelerate the market capitalization from the price perspective.

1.6 PLAN OF WORK

In order to achieve the above objectives the research has been divided into the following sections:

Section 2 – In this section an attempt has been made to understand the rise and growth of FIIs in India.

Section 3 – In this section an attempt has been made to understand the growth, importance or dominance of IPOs in the Indian Financial markets.

Section 4 - In this chapter the impact of FII inflows into Indian IPOs and its impact on the Indian stock market has been analysed.

Section 5 - In this chapter we tried to understand the impact of FII investment proportions in each stock on the listing performance of the respective stocks.

SECTION 2

RISE AND GROWTH OF FOREIGN INSTITUTIONAL INVESTORS.

Twentieth century saw Global equity market reforms and liberalization. The Indian Financial Market and the stock market underwent various positive reforms as well. Realising the need to integrate with the global economy and also recognising the fact that foreign investments are an important tool to development, India decided to deregulate its investment policies since 1991. From 1990-91 onwards the Government of India initiated many economic reforms and deregulations to accelerate economic growth and move towards globalization of the economy.

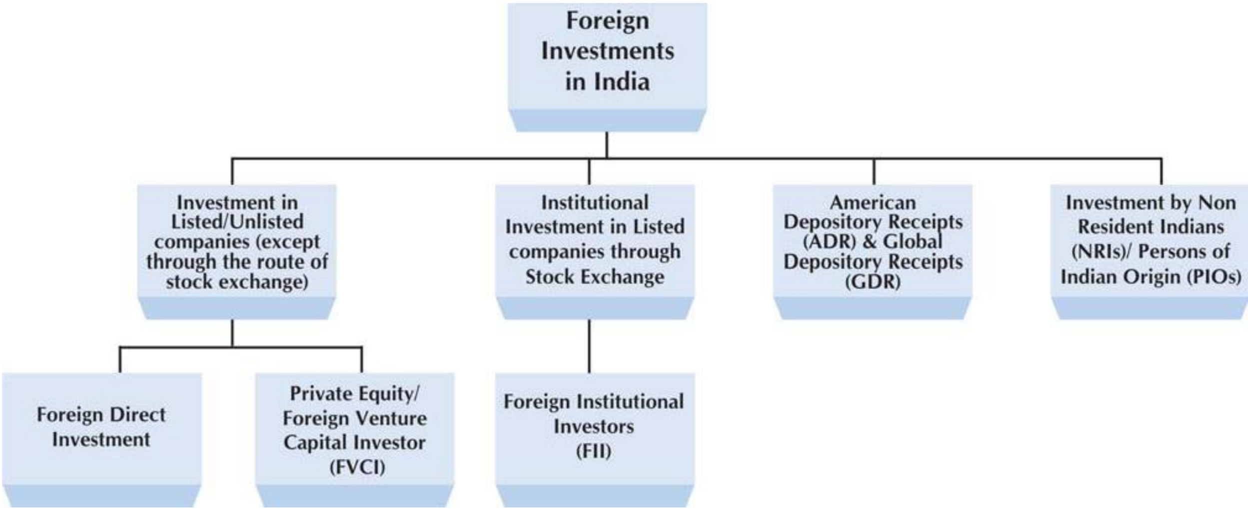
Till the 1980’s India managed its current account deficits mainly through debt flows and official development assistance and had an economy based on self-reliance and import substitution. However after the Gulf War India’s foreign exchange reserves were seriously affected, Balance of Payments position was badly hit until when IMF devalued rupee by 20 %.Thus came the era of liberalization and deregulation in the 1990’s with the removal of industrial licensing from several items, foreign investments invited in many sectors including foreign equity participation. Sectors like infrastructure, power generation, telecommunication, oil and natural gas exploration, services etc which earlier used to be state monopolized gradually got opened for the foreign investors. They were also welcome to invest in financial services, retail banking, life and general insurance. In this rise the Indian equity market received a boon in the name of Foreign Institutional Investors which were allowed to make portfolio investments in the growing Indian stock market.

FDI & FPI

According to **UNCTAD (1999)** portfolio investment involves transfer of financial assets by way of investment by resident individuals, enterprises and institutions in one country in securities of another country, either directly in the assets of the companies or indirectly through financial markets. The different components of FPI flows are Foreign Institutional Investors, Global/American Depository Receipts and offshore funds. **(Dua &Garg2013).** The Government, for the first time, permitted portfolio investments from abroad by foreign institutional investors in the Indian capital market in1992. “FDI is defined as cross-border investment by a resident entity in one economy with the objective of obtaining a lasting interest in an enterprise resident in another economy. The lasting interest implies the existence of a long-term relationship between the direct investor and the enterprise and a significant degree of influence by the direct investor on the management of the enterprise. Ownership of at least 10% of the voting power, representing the influence by the investor, is the basic criterion used." **(OECD Fact book 2013: Economic, Environmental and Social Statistics).**

Foreign Direct Investment (FDI) and Foreign Portfolio Investments (FPI) together constitute the Foreign Investment Inflows into India. Table 1 and Table 2 below explains the trend of FDI and FPI in the last decade which clearly shows that FPI flows had been much more volatile than FPI. Fig.1 shows the classification of Foreign Investment flows into India wherein the major constituents of FPIs are ADRs/GDRs, FIIs and Offshore Funds.

Fig 1: Classification of Foreign Investments in India.



Source: NSE Fact Book

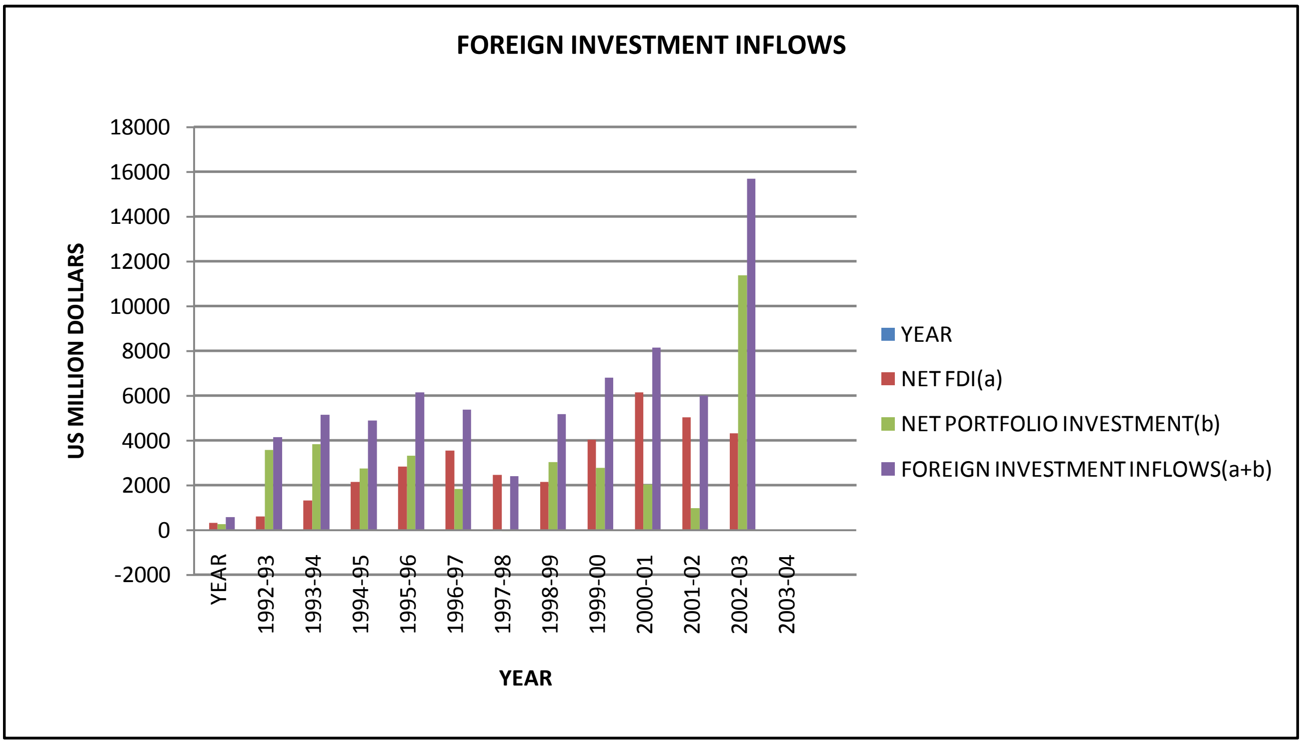
Table 1 Net FPI, Net FDI & Foreign Investment inflows of India (1992-2004)(US MN $)

YEAR	NET FDI(a)	NET PORTFOLIO INVESTMENT(b)	FOREIGN INVESTMENT INFLOWS(a+b)
1992-93	315	244	559
1993-94	586	3567	4153
1994-95	1314	3824	5138
1995-96	2144	2748	4892
1996-97	2821	3312	6133
1997-98	3557	1828	5385
1998-99	2462	-61	2401
1999-00	2155	3026	5181
2000-01	4029	2760	6789
2001-02	6130	2021	8151

2002-03	5035	979	6014
2003-04	4322	11377	15699

Source: RBI Website.

Fig 2: FDI & FPI Inflows from 1992-2004(US MN $)



Source: RBI Website.

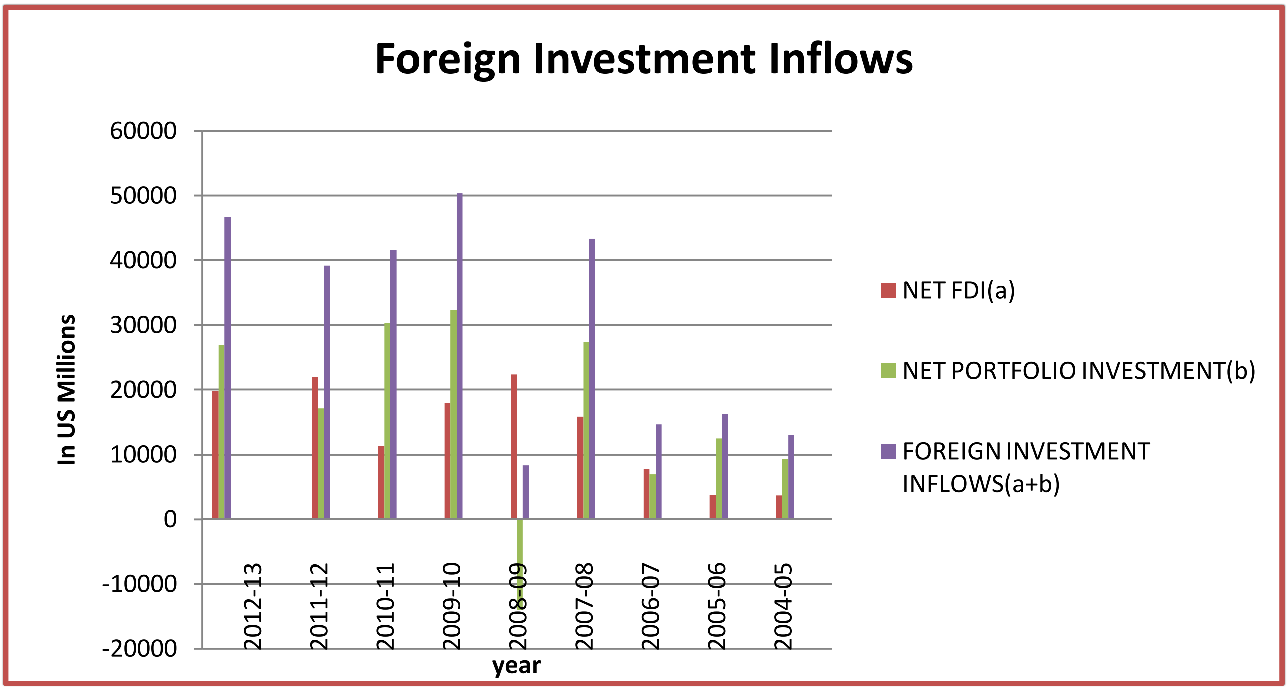
Table 2: Net FPI, Net FDI & Foreign Investment inflows of India (2005-2013) (US MN $)

YEAR	NET FDI(a)	NET PORTFOLIO INVESTMET(b)	FOREIGN INVESTMENT INFLOWS(a+b)
2012-13	19,819	26,891	46,710
2011-12	22,006	17,171	39,177

2010-11	11,305	30,292	41,597
2009-10	17,965	32,396	50,361
2008-09	22,343	-14,032	8,311
2007-08	15,891	27,434	43,325
2006-07	7,693	6,947	14,640
2005-06	3,769	12,492	16,261
2004-05	3,712	9,291	13,003

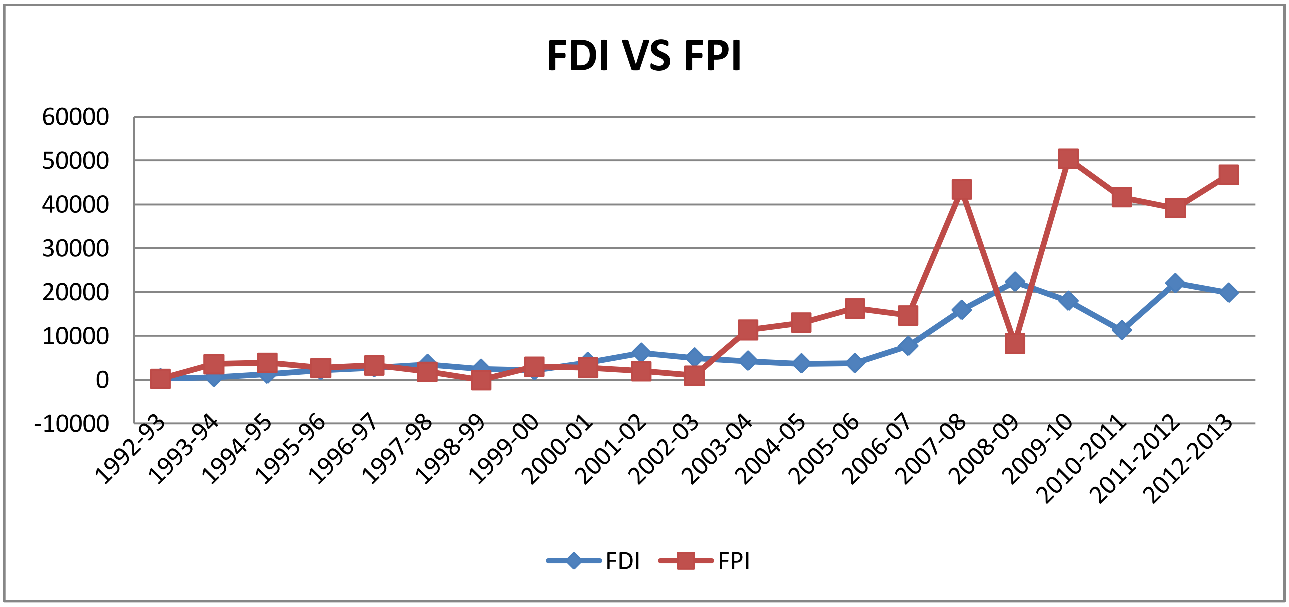
Source: RBI Website

Fig 3 FDI & FPI Inflows from 2005-2012(US MN $)



Source: RBI Website.

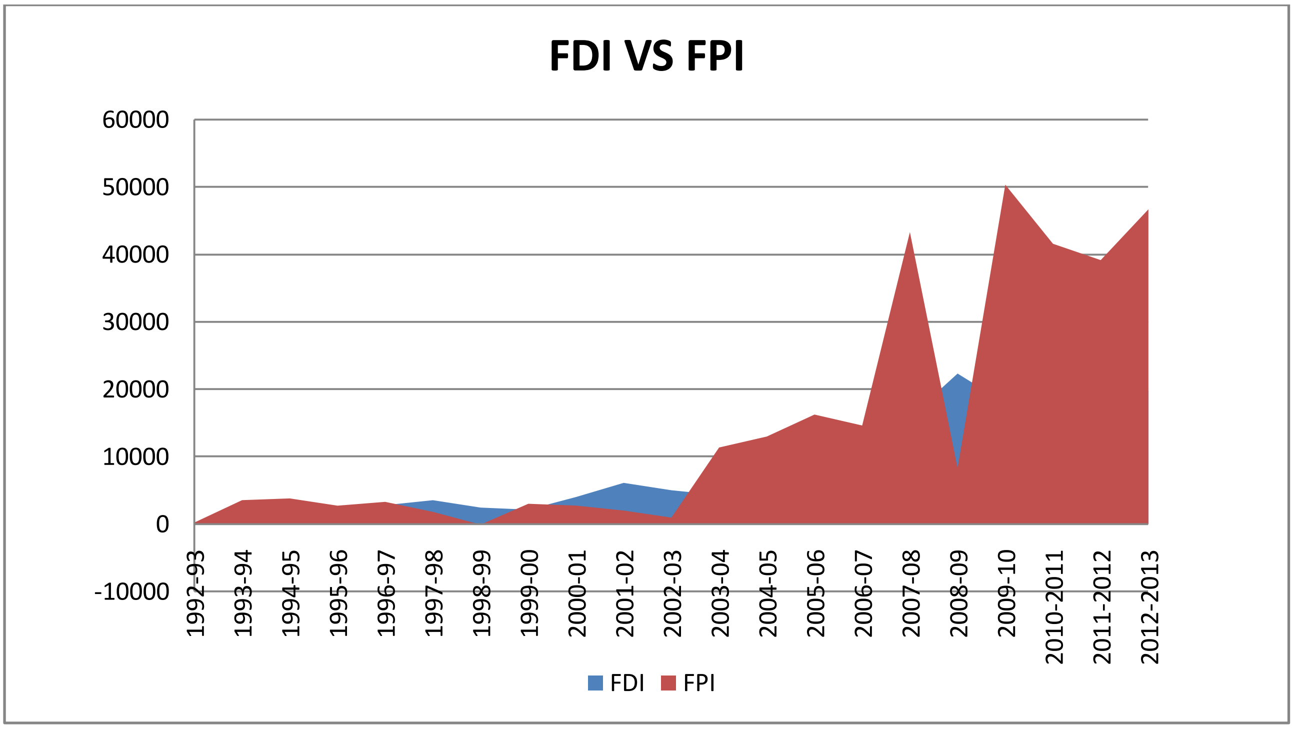
Fig 4 FDI vs. FPI (US MN $)



Source: RBI Website.

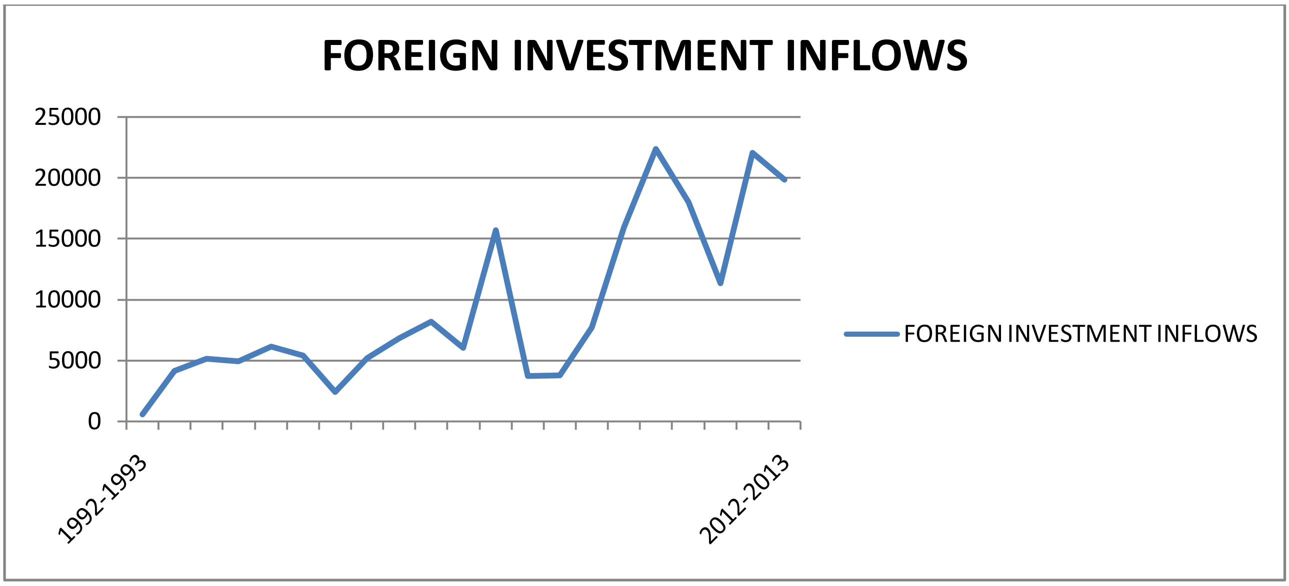
Post liberalization from the year 1992-93 we observe that the Net Portfolio Investments are gradually increasing. The FPI's have always been much more volatile than FDIs in total foreign investment inflows.(Fig 4) The money value of FPI in total is also much higher than that of FDI apart from a rising trend of the former.(Fig 5). In the last decade we observe that Foreign Portfolio Investments gradually increased until in the year 2008-09 (Fig 3) due to global meltdown there was negative portfolio investments. However the very fact of the reversal of FPI to high positive from negative within an year definitely indicates the confidence of foreign investors on emerging market economies like India.

Fig 5: Volume of FDI VS FPI(US MN $)



Source: RBI Website.

Fig 6 Trend of Total Foreign Investment Inflows (US MN $)



Source : RBI Website

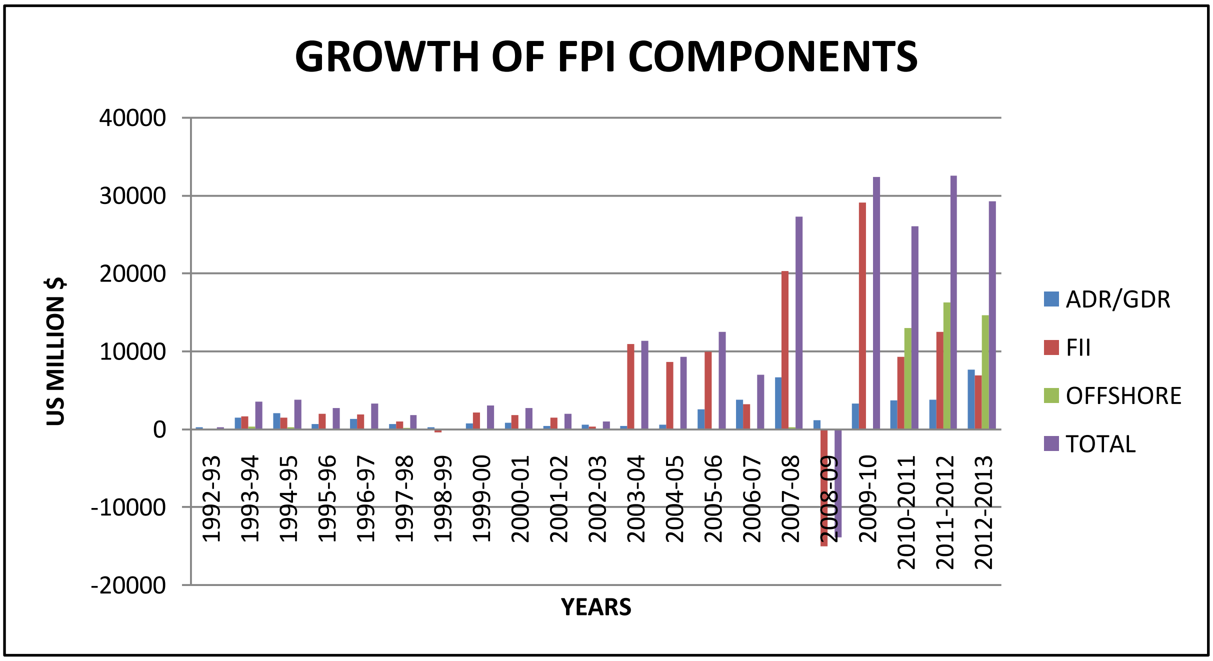
Over the years with increase in total foreign investment inflows (Fig 5) we find that the proportion of FPI is much higher than FDI in the total foreign investment inflows. The different components of FPI flows are Foreign Institutional Investors, Global/American Depository Receipts and offshore funds. American/Global Depository Receipts are instruments that are used by NRIs and foreign nationals for investing in Indian companies. Investment in Offshore financial funds i.e. those funds that are supposed to provide tax benefits to the investor are also included under the category of portfolio investment flows. One of the most important features of the development of stock market in India in the last 20 years has been the growing participation of FIIs.(Table 3)

T able 3 FPI flows (1992-2013) (US MN $)

	ADR/GDR	FII	OFFSHORE DERIVATIVES	TOTAL PORTFOLIO INVESTMENT INFLOWS
1992-93	240	1	3	244
1993-94	1520	1665	382	3567
1994-95	2082	1503	239	3824
1995-96	683	2009	56	2748
1996-97	1366	1926	20	3312
1997-98	645	979	204	1828
1998-99	270	-390	59	-61
1999-00	768	2135	123	3026
2000-01	831	1847	82	2760
2001-02	477	1505	39	2021
2002-03	600	377	2	979
2003-04	459	10918	0	11377
2004-05	613	8686	16	9315
2005-06	2552	9926	14	12492
2006-07	3776	3225	2	7003
2007-08	6645	20328	298	27271
2008-09	1162	-15017	0	-13855
2009-10	3328	29048	0	32376
2010-2011	3712	9291	13003	26006
2011-2012	3769	12492	16261	32522
2012-2013	7693	6947	14640	29280

Source: RBI Website.

Fig 7 Growth of FPI Components from 1992-2013



Source: RBI Website

SECTION - 3

THE DOMINANCE OF IPO IN THE FINANCIAL MARKET.

3.1. INTRODUCTION

Capital market can be divided into two segments viz. primary and secondary. The primary market is mainly used by issuers for raising fresh capital from the investors by making initial public offers or rights issues or offers for sale of equity or debt. The secondary market provides liquidity to these instruments, through trading and settlement on the stock exchanges. Capital market is, thus, important for raising funds for capital formation and investments and forms a very vital link for economic development of any country.

Primary Market: A market where the issuers access the prospective investors directly for funds required by them either for expansion or for meeting the working capital needs. This process is called disintermediation where the funds flow directly from investors to issuers.

IPO market in India has had its share of ups and downs over a period, for more than the last decade. It has seen a steep rise in the initial years of the post liberalization of Indian Economy. The growth observed during the first half of the 90s is mostly attributed to the financial liberalization of the economy. Capital market reforms like abolition of the office of controller of capital issues (CCI), constitution of SEBI under the new security and regulation act and relaxation in pricing of capital issues played an important role in such upsurge.

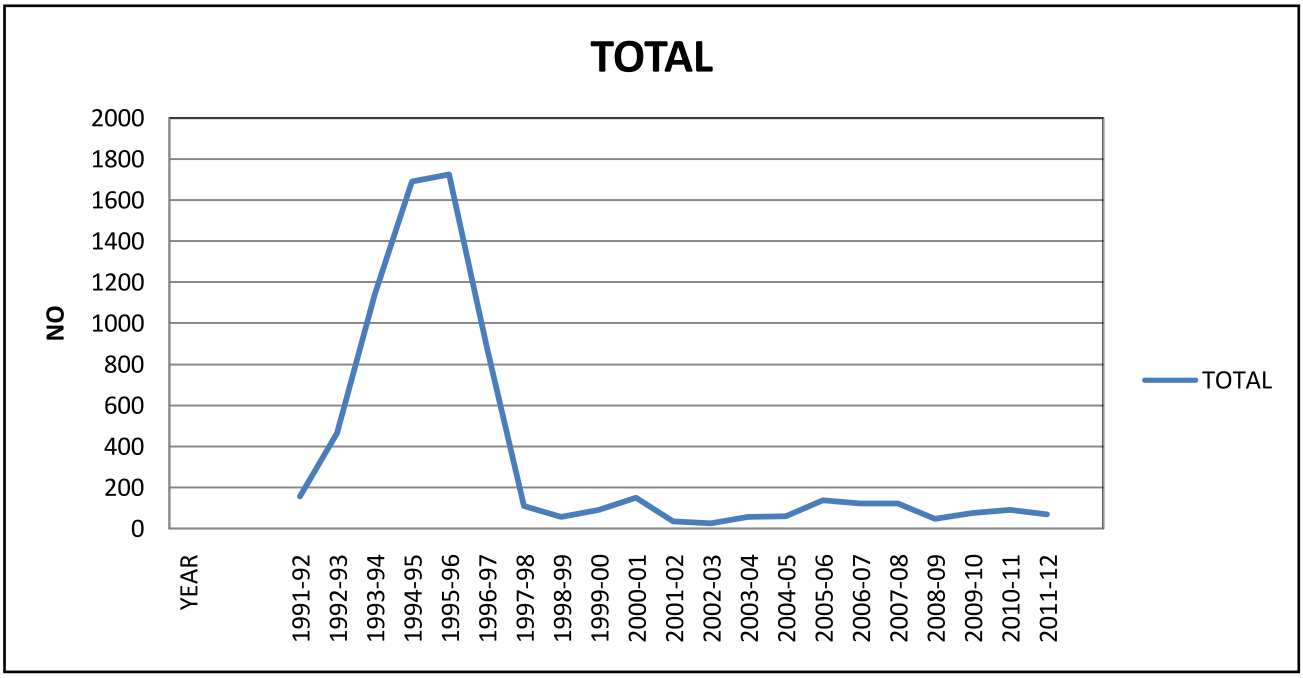
Table 4: Funds mobilised through IPO, FPO and Right Issues.(RS CRORES)

	TOTAL		IPO		RIGHTS		FPO	
YEAR	NO OF PRIMARY ISSUES	AMOUNT	NO	AMOUNT	NO	AMOUNT	NO	AMOUNT
		RS CRORES		RS CRORES				RS CRORES
1993-94	1143	24372	692	7864	370	8923	81	7585
1994-95	1692	27633	1239	16572	350	6588	103	4473
1995-96	1725	20804	1357	10924	299	6564	69	3316
1996-97	884	14284	717	5959	131	2719	36	5606
1997-98	111	4570	52	1048	49	1708	10	1814

1998-99	58	5587	18.00	404	26	568	14	4615
1999-00	93	7817	51	2719	28	1560	14	3538
2000-01	151	6108	114	2722	27	729	10	2657
2001-02	35	7543	7	1202	15	1041	13	5300
2002-03	26	4070	6	1039	12	431	8	2600
2003-04	57	23272	21	3434	22	1007	14	18831
2004-05	60	28256	23	13749	26	3616	11	10891
2005-06	139	27382	79	10936	36	4088	24	12358
2006-07	124	33508	77	28504	39	3711	8	1293
2007-08	124	87029	85	42595	32	32518	7	11916
2008-09	47	16220	21	2082	25	12638	1	1500
2009-10	76	57555	39	24696	29	8319	8	24540
2010-11	91	67609	53	35559	23	9503	15	22547
2011-12	71	48468	54	41515	16	2375	1	4578

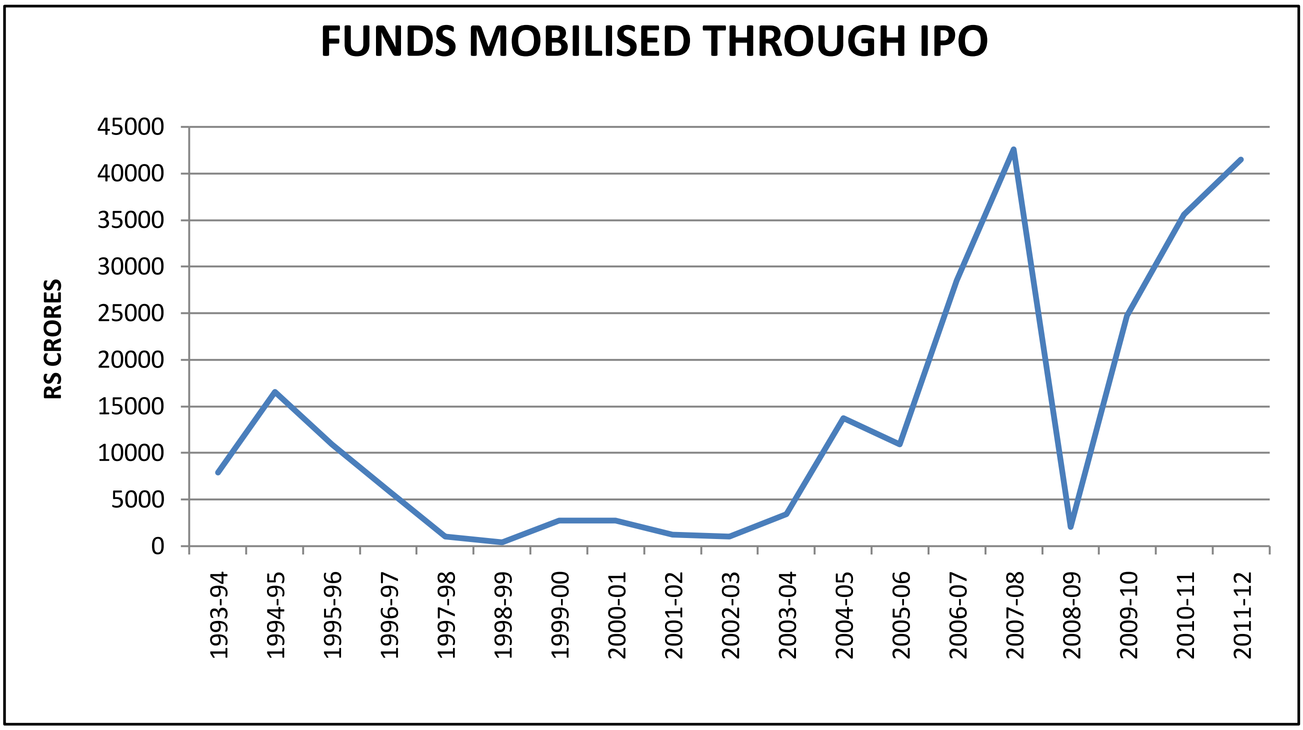
Source: SEBI Hand book 2012

Fig 8: Growth in the number of IPOs from 1992-2012



Source: SEBI Hand book 2012

Fig 9 Funds Mobilised through IPOs from 1993 to 2012.(RS CRORES)

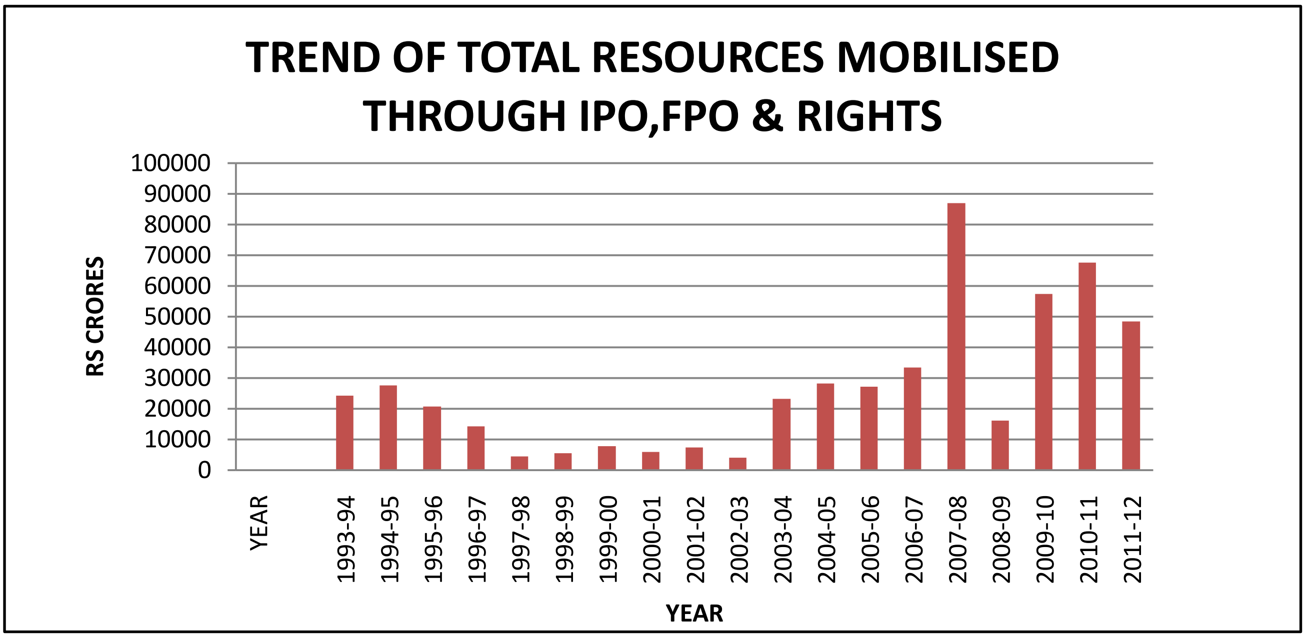


Source: SEBI Hand book 2012

Fig 8 shows that the number of IPOs reached at its peak in the early 1990's after which it gradually started falling and from 1998-99 onwards the growth in the number of IPO's shows a little and almost constant growth. This implies that the corporate coming up with fresh issue has drastically declined in the last decade which is also evident from table 4 above.

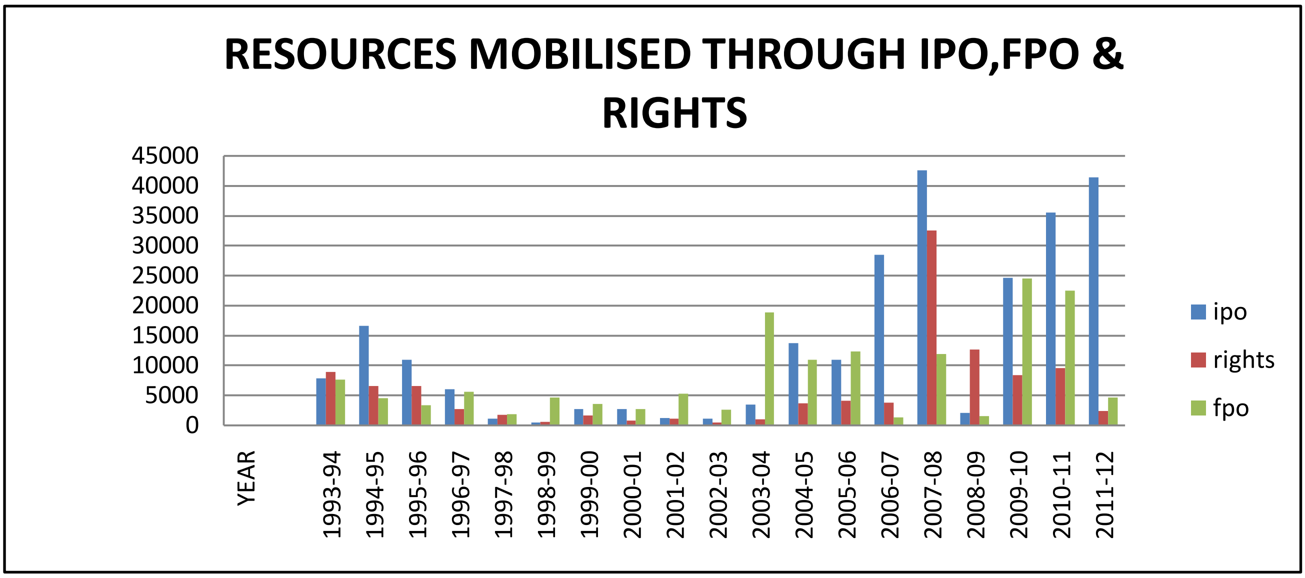
However the funds mobilized through IPO(Fig 9) do not follow the same pattern. The funds mobilized reached a peak in 2006-2007 period until from 2008 onwards primarily due to Global meltdown we see a steep decline in the amount of capital formation through IPO's.

Fig 10 Total Resources mobilised through IPO, FPO & Rights. .(RS CRORES)



Source: SEBI Hand book 2012

Fig 11 Component wise resources mobilised through IPO, FPO & Rights. .(RS CRORES)



Source: SEBI Hand book 2012

Thus we see that the primary market of India went through various ups and downs over the last decades. However it is imperative to understand that it is the primary market of India in which the corporate come up with public issues before they can actually be launched in the secondary market. It is the Primary market which acts as the focal point for capital formation for the companies for their various requirements of funds. As we know that amongst the various investor categories by now the Foreign Institutional investors have gained increased importance due to the cross border capital flows gaining importance world over. Though the Foreign Institutional investors have gained much focus as market movers for the secondary market it is of utmost importance to understand the role they play in the primary market of India.

3.2 RESEARCH HYPOTHESES AND METHODOLOGY

Thus we first proceed to understand the impact of IPO on the financial markets. The financial market has been represented by NSE market capitalisation and Nifty.

RH_1=Market capitalisation is affected by IPO volumes.

RH_2=Nifty is affected by IPO volumes.

The study expects to support the view that IPO volumes affect market capitalisation and Nifty.

TIME PERIOD

We have considered a time period of 2006 to 2011 in our study. We have taken all IPO's that have come up during this period and have been listed on the NSE(National Stock Exchange).We have taken calendar year as the basis of our study. The number of IPO's which were listed in the respective years were as follows:

YEAR	NO OF IPOS
2006	67
2007	94
2008	35
2009	15
2010	62
2011	34

Thus we analyzed these 308 IPO's for the purpose of our study.

DATA

For the purpose of explaining the data collection process we have explained the data collection process of the year 2006 in this section and data collection for all other years have been done in similar way. For this section we collected data of the following:

a. Monthly IPO Volumes.
b. Quarterly GDP values at factor cost.
c. Closing Monthly Nifty Values.
d. Closing monthly market capitalization.

We have used GDP as the control variable as a result of which all monthly data were reduced to quarterly data as GDP is only available on a quarterly basis.

a. <u>IPO VOLUMES</u>

The IPO volumes are the monthly total of all IPO volumes being issued in that month which we have collected from the NSE website. It is the number of shares issued multiplied by the issue price which denotes the size or volume of the IPO.

Thus from the above table we obtain the total IPO volumes of each month by totaling the individual IPO volumes of the companies that came up with their issues in each month. In this way we obtained the monthly IPO volumes from 2006 to 2011 i.e. for a period of 72 months.

Table 5: Monthly IPO Volumes for 2006 (Amount in Rs MN)

JANUARY		**FEBRUARY**		**MARCH**	
NAME OF THE SCRIPT	**AMOUNT**	**NAME OF THE SCRIPT**	**AMOUNT**	**NAME OF THE SCRIPT**	**AMOUNT**
BARTRONICS INDIA LTD.	450	GVK POWER & INFRASTRUCTURE LTD	2565.42	VISA STEEL LTD	1710
CELEBRITY FASHIONS LTD	819	INOX LEISURE LIMITED	1980	NITCO TILES LTD	1680
EDUCOMP SOLUTIONS LTD.	500	JAGRAN PRAKASHAN LIMITED	3212	J.K CEMENT LIMITED	2960
PUNJ LOYD LTD	6421.05	GUJARAT STATE PETRONET LTD	3726	MAHINDRA AND MAHINDRA FINANCIAL SERVICES	4000
TULIP IT SERVICES LTD.	1080	ENTERTAINMENT NETWORK INDIA LTD	2138.4	B.L. KASHYAP AND SONS LTD	1883.75
PVR LTD DEMAT	1665	ROYAL ORCHIDS HOTELS LIMITED	1125.3	PRATIBHA INDUSTRIES LTD.	514.2
		NITIN SPINNERS LTD.	466.67	K SERA SERA PRODUCTIONS LTD	340
				GITANJALI GEMS LTD.	3315
				SADBHAV	536.5

				ENGINEERING LTD.	
				INDO TECH TRANSFORMERS LTDDM	596
				SUNIL HITECH ENGINEERS LTD.	346.5
TOTAL	**10935.06**	**TOTAL**	**15213.79**	**TOTAL**	**17881.95**
APRIL		MAY		JUNE	
NAME OF THE SCRIPT	AMOUNT	NAME OF THE SCRIPT	AMOUNT	NAME OF THE SCRIPT	AMOUNT
SUN TV LTD DM	6027.87	PLETHICO PHARMACEUTICALS LTD.	1178.56	ALLCARGO GLOBAL LOGISTICS PRIVATE LTD	1403.32
EMKAY SHARE AND STOCK BROKERS LTD.	750	LOKESH MACHINES LTD.	420	PRIME FOCUS LTD.	1000
GODAWARI POWER AND ISPAT LTD	704.3	KAMDHENU ISPAT LTD	228	DECCAN AVAITION LTD	3632.8
R SYSTEMS INTERNATIONAL LIMITED	1102.08			UNITY INFRAPROJECTS LTD.	2324.03
KEWAL KIRAN CLOTHING LTD.	806				
UTTAM SUGAR MILLS LTD.	1360				
ADHUNIK	999.99				

METALIKS LTD.					
SOLAR EXPLOSIVES LTD.	836				
GALLANTT METAL LTD.	85				
MALU PAPER MILLS LTD	200				
SHIVALIK GLOBAL, SPL GLOBAL LTD	540				
ROHIT FERRO-TECH LIMITED.	232				
TOTAL	**13643.24**	**TOTAL**	**1826.56**	**TOTAL**	**8360.15**
JULY		AUGUST		SEPTEMBER	
NAME OF THE SCRIPT	AMOUNT	NAME OF THE SCRIPT	AMOUNT	NAME OF THE SCRIPT	AMOUNT
		GMR INFRASTRUCTURE LIMITED	8008.77	HOV SERVICES LIMITED	810
		TECH MAHINDRA LTD.	4652.29	ATLANTA LIMITED	645
				ACTION CONSTRUCTION EQP LTD	2990
				VOLTAMP TRANSFORMERS LIMITED	1684.93

TOTAL	**NIL**	**TOTAL**	**12661.06**	**TOTAL**	**6129.93**
OCTOBER		NOVEMBER		DECEMBER	
NAME OF THE SCRIPT	AMOUNT	NAME OF THE SCRIPT	AMOUNT	NAME OF THE SCRIPT	AMOUNT
GLOBAL VECTRA HELICORP LTD	647.5	PARSVNATH DEVELOPERS LTD.	9971.4	ESS DEE ALUMINIUM LIMITED	1566
DEVELPMENT CREDIT BANK	1859	LANCO INFRATECH LTD.	10673.37	XL TELECOM LTD.	593.52
ACCEL FRONTLINE LIMITED	422.69	INFO EDGE (INDIA) LIMITED	1703.63	L.T.OVERSEAS LTD.	393.99
HANUNG TOYS AND TEXTILES LTD.	902.5			SOBHA DEVELOPERS LTD.	5691.73
JHS SVENDGAARD LABS LTD	359.6			BLUE BIRD INDIA LIMITED	921.38
FIEM INDUSTRIES LIMITED	561.7			NISSAN COPPER LIMITED	250
GWALIOR CHEMICAL INDUS LTD	800			RUCHIRA PAPER MILLS LTD	235
GAYATRI PROJECTS LIMITED	408.9				
TOTAL	**5961.89**	**TOTAL**	**22348.4**	**TOTAL**	**9651.62**

Source: NSE Website

Table 6: Quarterly IPO Volumes from 2006-2011(RSMN)

QUARTER ENDING	IPO VOLUME
MARCH'06	44030.79
JUNE'06	23829.94
SEP'06	18790.98
DECEMBER'06	37961.91

QUARTER ENDING	IPO VOLUME
MARCH'08	153821.6
JUNE'08	11394.48
SEP'08	16269.88
DECEMBER'08	501.97

QUARTER ENDING	IPO VOLUME
MARCH'10	80670.3
JUNE'10	48103
SEP'10	35003
DECEMBER'11	240280

QUARTER ENDING	IPO VOLUME
MARCH'07	130650.2
JUNE'07	17366.73
SEP'07	256808.2
DECEMBER'07	83752.69

QUARTER ENDING	IPO VOLUME
MARCH'09	238.43
JUNE'09	0
SEP'09	124408.7
DECEMBER'09	31756.75

QUARTER ENDING	IPO VOLUME
MARCH'11	10469
JUNE'11	22076
SEP'11	21973
DECEMBER'11	4665.75

Source: NSE Website

b. <u>Gross Domestic Product (GDP)</u>

Table 7: GDP at factor cost from 2006-2011(RS BN)

QUARTER ENDING	GDP
MARCH'06	8845.78
JUNE'06	8308.76
SEP'06	8258.33
DECEMBER'06	9362.37

QUARTER ENDING	GDP
MARCH'08	10546.85
JUNE'08	10009.47
SEP'08	9816.00
DECEMBER'08	10849.93

QUARTER ENDING	GDP
MARCH'10	12151.94
JUNE'10	11606.11
SEP'10	11652.64
DECEMBER'11	12751.40

QUARTER ENDING	GDP
MARCH'07	9714.18
JUNE'07	9116.02
SEP'07	9045.29
DECEMBER'07	10258.18

QUARTER ENDING	GDP
MARCH'09	10911.35
JUNE'09	10601.28
SEP'09	10725.06
DECEMBER'09	11682.45

QUARTER ENDING	GDP
MARCH'11	13359.89
JUNE'11	12474.96
SEP'11	12411.06
DECEMBER'11	13512.52

Source: RBI Website.

c. Closing Monthly Nifty Values

Table8: Monthly closing of Nifty values from 2006-2011.

MONTHS	NIFTY
Jan'06	2892.7
Feb'06	3019.3
Mar'06	3236.4
Apr'06	3494.1
May'06	3437.4
Jun'06	2914.9
Jul'06	3092.1
Aug'06	3305.6
Sep'06	3492.1
Oct'06	3649.4
Nov'06	3868.6
Dec'06	3910.2

MONTHS	NIFTY
Jan'08	5756.4
Feb'08	5201.6
Mar'08	4769.5
Apr'08	4901.9
May'08	5028.7
Jun'08	4463.8
Jul'08	4124.6
Aug'08	4417.1
Sep'08	4206.7
Oct'08	3210.2
Nov'08	2834.8
Dec'08	2895.8

MONTHS	NIFTY
Jan'10	5156.2
Feb'10	4839.6
Mar'10	5178.1
Apr'10	5294.8
May'10	5053
Jun'10	5187.8
Jul'10	5359.7
Aug'10	5457.2
Sep'10	5811.5
Oct,10	6096.1
Nov'10	6055.3
Dec'10	5971.3

MONTHS	NIFTY
Jan'07	4037.1
Feb'07	4083.7
Mar'07	3731.1
Apr'07	3947.3
May'07	4184.4
Jun'07	4222.2
Jul'07	4474.2
Aug'07	4301.4
Sep'07	4659.9
Oct'07	5456.6
Nov'07	5748.6
Dec'07	5963.6

MONTHS	NIFTY
Jan'09	2854.4
Feb'09	2819.2
Mar'09	2802.3
Apr'09	3359.8
May'09	3958
Jun'09	4436.4
Jul'09	4343
Aug'09	4571.1
Sep'09	4859.3
Oct'09	4994.1
Nov'09	4953.5
Dec'09	5099.7

MONTHS	NIFTY
Jan'11	5782.7
Feb'11	5400.9
Mar'11	5538.4
Apr'11	5839.1
May'11	5492.2
Jun'11	5472.6
Jul'11	5596.6
Aug'11	5076.7
Sep'11	5015.6
Oct'11	5060
Nov'11	5004.3
Dec'11	4782.4

Source: SEBI Handbook 2012

For the purpose of our analysis we have considered the quarterly closing values of the Nifty index.

d. Market Capitalisation

We considered the Market capitalization of the NSE by considering data from the SEBI website. The monthly closing values of market capitalization of NSE have been considered for the purpose.

Table 9: Monthly closing values of Market capitalisation from 2006-2011. **(RS MN)**

YEAR/MONTH	**2006**	**2007**	**2008**	**2009**	**2010**	**2011**
JAN	243439.5	357148.7	529538.7	279870.7	578296.5	644149.1
FEB	251208.3	329693.1	541994.2	267562.2	575530.5	619596.7
MARCH	281320.1	336735	485812.2	289619.4	600917.3	670261.6
APRIL	299020	365036.8	544278	337502.5	611785.8	675361.4
MAY	261263.9	389807.8	509887.3	456457.2	593257.8	656974.3
JUNE	252465.9	397838.1	410365.1	443259.6	622913.6	657474.3
JULY	251426.1	431757.1	443242.7	481645.9	634012	646223.8
AUG	277740.1	429699.4	447246.1	497580	639341.8	592168.4
SEP	299413.2	488656.1	390018.5	535388	695853.4	582033.4
OCT	313831.9	572222.7	282038.8	502483	705509.4	610189.1
NOV	337365.2	587674.2	265328.1	543008.8	689491.2	605925.8
DEC	342623.6	654327.2	291676.8	569963.7	713931	569554.7

Source: SEBI Handbook 2012

For the purpose of our analysis we considered the closing values at the end of each quarter.

ANLYTICAL TOOLS AND TECHNIQUES

In order to analyze the collected data the statistical tools such as Correlation, Multiple regression OLS model is used. The multiple regression analysis is a statistical technique used to evaluate the effects of two or more independent variables on a single dependent variable. In this model we try to establish the amount of dependency of Market capitalisation on IPO volume and GDP. Thus in our model Market capitalisation has been considered as the dependent variable and IPO and GDP have been considered as independent variables.

To deduce the above hypothesis we formulate a multiple linear regression analysis, wherein the response variable Y_I is a function of k explanatory variables $x_{i,1}, \ldots, x_{i,k}$. This relationship is straight-line and in its basic form it can be written as

$$Y_i = \beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2} + \cdots + \beta_k x_{i,k} + \varepsilon_i,$$

where the random errors $\varepsilon_i,\ i = 1, \ldots, n,$ are independent normally distributed random variables with zero mean and constant variance σ^2

The definition of a multiple linear regression model is that mean of the response variable,

$$\mathbb{E}[Y_i] = \beta_0 + \beta_1 x_{i,1} + \beta_2 x_{i,2} + \cdots + \beta_k x_{i,k},$$

is a linear function of the regression parameters $\beta_0, \beta_1, \ldots, \beta_k$

3.3 DATA ANALYSIS AND INTERPRETATION

RESULTS EXPLAINING MODEL

Table 10 Descriptive Statistics of Market capitalisation, IPO Volumes, GDP and Nifty.

	N	Minimum	Maximum	Mean	Std. Deviation
MCAPQT	24	2524659	7139310	4826009.	1503615
GDPQTR	24	8258330	13512520	10748825	1542118
IPOQRT	24		256808	58950	72428
NIFTYCL	24	2802	5971	4525	968

Where N = 24 quarters.

Model 1

MARKETCAPITALISATION =B + b_1IPO$_i$ +b_2 GDP$_i$ +e_i .... (1)

Where,

MARKET CAPITALISATION$_i$ is the quarterly closing values of the NSE.

GDP$_i$ is the quarterly values of gross domestic product at factor cost of the country.

IPO$_i$ is the quarterly total volumes of the IPOs.

e_i is the error term are independent normally distributed random variables with zero mean and constant variance σ^2, i=1..n., .n=24.

b_1 and b_2 are the unknown model parameters

Table 11 Regression Results of IPO and GDP on Market capitalisation

DEPENDENT VARIABLE - MARKET CAPITALISATION

Independent Variables	IPO	GDP	
Standardized beta coefficient	.309	.785	
t ratio	2.526	6.424	
Significance	.020	.000	
Collinearity Tolerance VIF	 .997 1.003	 .997 1.003	
R Square			.687
Adj R Square			.657
Durbin Watson			1.687
F Statistic			23.069
Significance of F statistic			.000

Fig 12 Normal P-P Plot of regression standardised residual of IPO & GDP on market capitalisation.

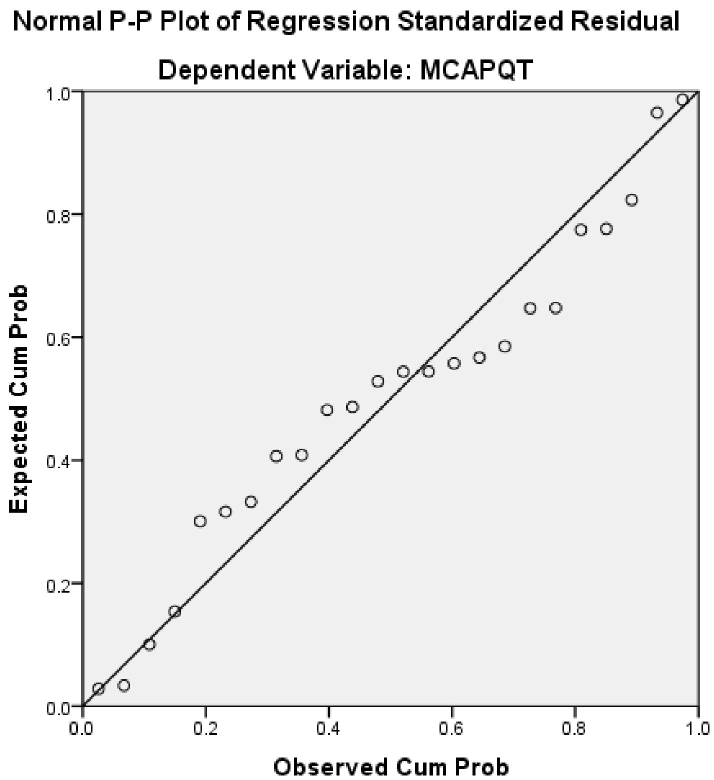
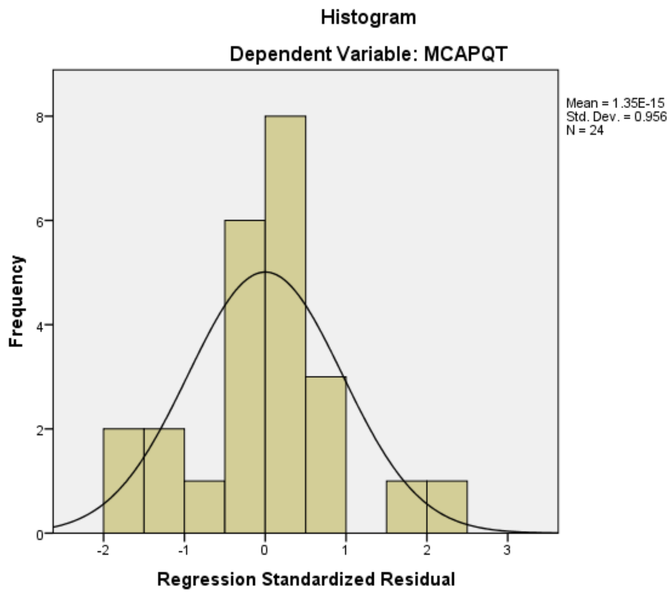


Fig 13 Histogram of regression standardised residual of IPO & GDP on market capitalisation



INTERPRETATION OF SUMMARY STATISTICS

From Table 11 we see that R^2 is .687 and the adjusted R^2 is .657.This means that the explanatory variables together account for explaining around 69% of the variation in the response variable market capitalisation. The difference for the final model is very small i.e .687-.657 = .030 .This shrinkage means that if the model were derived from the population rather than a sample it would account for approximately .030 less variance in the outcome.

The F ratio represents the ratio of the improvement in prediction that results from fitting the model relative to the inaccuracy that still exists in the model. In this case the F ratio is 23.069 which is satisfactory and is significant at 1 % level.

The Durbin Watson test statistic to check for serial correlation of the random errors is 1.687 and being closer to 2 implies that the error terms are independent.

We also note from the VIF and Tolerance statistic that multicollineariy is not a serious problem in our model as the tolerance is far above .1 and VIF is far below 10.

Moreover the P-P plot for checking normality of the residual shows that the residual is almost following a normal distribution.

INTERPRETATION OF REGRESSION PARAMETERS

The b- values tells us about the relationship between market capitalisation and each of the independent predictors IPO and GDP. The t statistic being significant implies that the b values are significantly different from zero as the beta value of IPO is significant at 1% level and beta value of GDP is significant at 1% level. In other words it implies that the predictors are making significant contribution to the model.

Model 2

$NIFTY_i = B + b_1 IPO_i + b_2\ GDP_i + e_i$ .... (1)

Where,

$NIFTY_i$ is the quarterly closing values of the NIFTY.

GDP_i is the quarterly values of gross domestic product at factor cost of the country.

IPO_i is the quarterly total volumes of the IPOs.

e_i is the error term are independent normally distributed random variables with zero mean and constant variance σ^2, i=1..n.

b_1 and b_2 are the unknown model parameters.

i=1….n n=24

Table 12 Regression results of IPO and GDP on Nifty

DEPENDENT VARIABLE - NIFTY

Independent Variables	IPO	GDP	
Standardized beta coefficient	.354	.656	
t ratio	2.368	4.389	
significance	.028	.000	
Collinearity Tolerance	.997	.997	

VIF	1.003	1.003	
R Square			.532
Adj R Square			.488
Durbin Watson			1.771
F Statistic			11.944
Significance of F statistic			.000

Fig 14 Histogram of regression standardised residual of IPO & GDP on Nifty

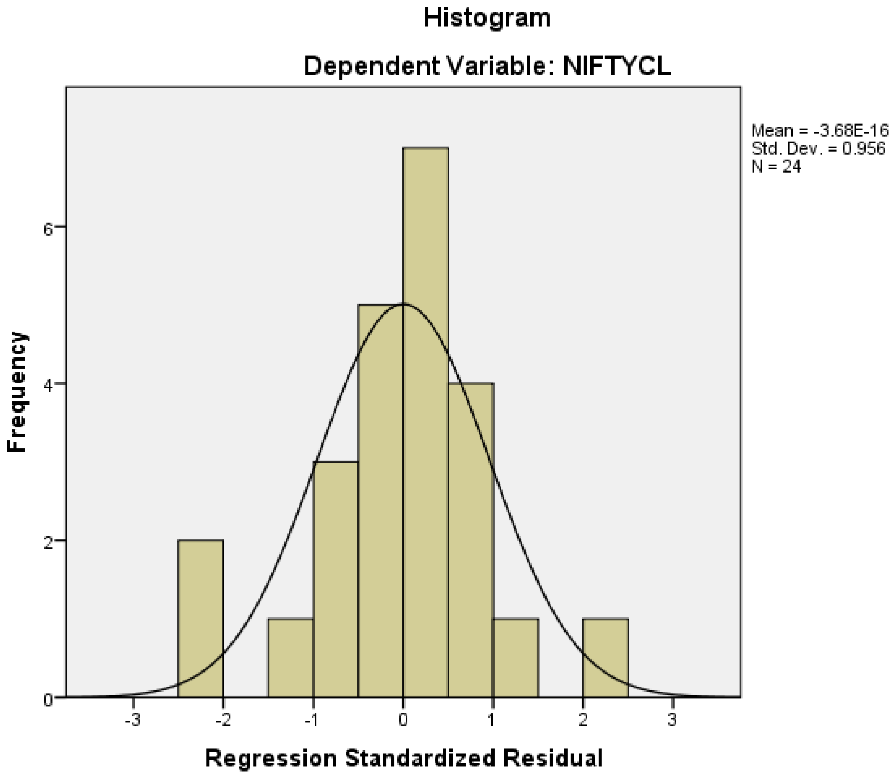
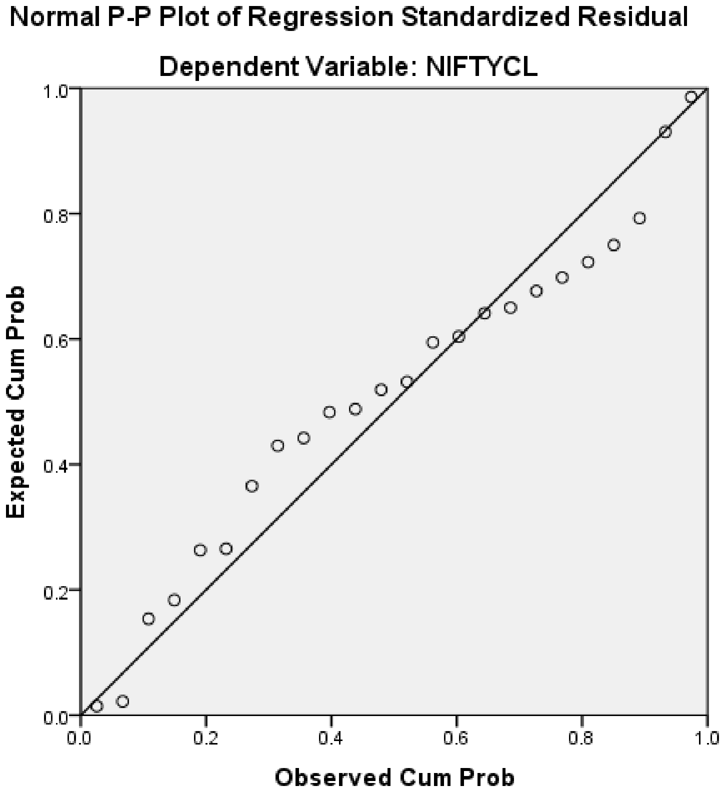


Fig 15 Normal P-P Plot of regression standardised residual of IPO & GDP on Nifty.



INTERPRETATION OF SUMMARY STATISTICS

From Table 3 we see that R^2 is .687 and the adjusted R^2 is .532.This means that the explanatory variables together account for explaining around 53% of the variation in the response variable NIFTY. The difference for the final model is very small i.e. .532-.488 = .040 .This shrinkage means that if the model were derived from the population rather than a sample it would account for approximately .040 less variance in the outcome.

The F ratio represents the ratio of the improvement in prediction that results from fitting the model relative to the inaccuracy that still exists in the model. In this case the F ratio is 11.944 which is satisfactory and is significant at 1 % level.

The Durbin Watson test statistic to check for serial correlation of the random errors is 1.771 and being closer to 2 implies that the error terms are independent.

We also note from the VIF and Tolerance statistic that multicollineariy is not a serious problem in our model as the tolerance is far above .1 and VIF is far below 10.

Moreover the P-P plot for checking normality of the residual shows that the residual is almost following a normal distribution.

INTERPRETATION OF REGRESSION PARAMETERS

The b- values tells us about the relationship between NIFTY and each of the independent predictors IPO andGDP. The t statistic being significant implies that the b values are significantly different from zero as the beta value of IPO is significant at 1% level and beta value of GDP is significant at 1% level. In other words it implies that the predictors are making significant contribution to the model.. Thus we reject our null Hypothesis and accept the alternate hypothesis at 1 % a levelsignificance and arrive at the conclusion that both IPO andGDP influence NIFTY.

SECTION 4

IMPACT OF FII INVESTMENTS INTO IPO ON THE INDIAN STOCK MARKET.

4.1 INTRODUCTION

As already evident from the last section that Nifty positively affects the IPO volumes of our country. On the other hand companies raise a major portion of their funds through institutional investors – the Domestic Institutional Investors and the Foreign Institutional investors. It is presumed that when the Financial Markets reflect positivity the FII inflows into IPOs will be higher. The stock market indices are significant in explaining market sentiments and prospects. Thus we might primarily assume that there may lay a fundamental linkage between the Primary market, Secondary market and the FII inflows into the primary market. Researches in the past have shown numerous such attempts in explaining market sentiments through net FII inflows into the secondary market. But in this section we will try to analyse whether the FII inflows into IPO have any significant impact on the NIFTY.

4.2 RESEARCH HYPOTHESES AND METHODOLOGY

Hypothesis

Thus based on the above objectives we frame the following research hypothesis:

RH_{1} = FII inflows into IPOs have impact on NIFTY ……

RH_2= FII inflows into IPO have impact on market capitalisation.

The study expects to support the view that FII inflows into IPO and GDP have influence on NIFTY and market capitalisation.

Research Methodology

The time period and analytical tools used in the previous section remains the same for this section as well.

Data

In this section we have used the following data of the following variables for our analysis:

a. FII inflows into IPO.

b. Nifty values.

c. GDP

d. NSE market capitalisation.

For the purpose of explaining the data collection process we have explained the data collection process of the year 2006 in this section and data collection for all other years have been done in similar way.

FII INFLOWS INTO IPO

Firstly we had to find out the FII inflows into each of these IPOs. The data for this purpose was taken from the FII statistics section of trade wise equity data of FII's from the website of Securities Exchange Board of India. Every IPO that got listed during this period had their date wise transactions listed in the SEBI (Securities Exchange Board of India) website which also includes daily transactions with the FII's. This was the most unique part of the study wherein we attempted to compile the total of FII inflows into each of these IPO's.

Table 13 FII Inflows into IPOs of 2006.(RS MN)

YEAR 2006

JANUARY		**FEBRUARY**		**MARCH**	
NAME OF THE SCRIPT	**AMOUNT**	**NAME OF THE SCRIPT**	**AMOUNT**	**NAME OF THE SCRIPT**	**AMOUNT**
BARTRONICS INDIA LTD.	98.61	GVK POWER & INFRASTRUCTURE LTD	663.52	VISA STEEL LTD	744.38
CELEBRITY FASHIONS LTD	202.94	INOX LEISURE LIMITED	524.72	NITCO TILES LTD	461.55
EDUCOMP	94.75	JAGRAN	659.77	J.K CEMENT	1190.95

SOLUTIONS LTD.		PRAKASHAN LIMITED		LIMITED	
PUNJ LOYD LTD	2776.47	GUJARAT STATE PETRONET LTD	553.15	MAHINDRA AND MAHINDRA FINANCIAL SERVICES	1913.71
TULIP IT SERVICES LTD.	209.79	ENTERTAINMENT NETWORK INDIA LTD	608.97	B.L. KASHYAP AND SONS LTD	631.94
PVR LTD DEMAT	368.18	ROYAL ORCHIDS HOTELS LIMITED	305.75	PRATIBHA INDUSTRIES LTD.	110.93
		NITIN SPINNERS LTD.	90.20	K SERA SERA PRODUCTIONS LTD	137.83
				GITANJALI GEMS LTD.	1082.76
				SADBHAV ENGINEERING LTD.	132.32
				INDO TECH TRANSFORME RS LTDDM	48.12
				SUNIL HITECH ENGINEERS LTD.	19.97
TOTAL	3750.73	TOTAL	3406.08	TOTAL	6474.46
APRIL		**MAY**		**JUNE**	
NAME OF THE SCRIPT	**AMOUNT**	**NAME OF THE SCRIPT**	**AMOUNT**	**NAME OF THE SCRIPT**	**AMOUNT**
SUN TV LTD DM	2580.06	PLETHICO PHARMACEUTICALS LTD.	429.77	ALLCARGO GLOBAL LOGISTICS	571.47

				PRIVATE LTD	
EMKAY SHARE AND STOCK BROKERS LTD.	235.99	LOKESH MACHINES LTD.	105.51	PRIME FOCUS LTD.	354.87
GODAWARI POWER AND ISPAT LTD	95.37	KAMDHENU ISPAT LTD	0.00	Deccan Aviation Limited	829.28
R SYSTEMS INTERNATIONAL LIMITED	357.35			UNITY INFRAPROJECTS LTD.	858.64
KEWAL KIRAN CLOTHING LTD.	282.69				
UTTAM SUGAR MILLS LTD.	588.00				
ADHUNIK METALIKS LTD.	303.37				
SOLAR EXPLOSIVES LTD.	175.39				
GALLANTT METAL LTD.	7.32				
MALU PAPER MILLS LTD	0.00				
SHIVALIK GLOBAL, SPL GLOBAL LTD	0.00				
ROHIT FERRO-TECH LIMITED.	106.70				
TOTAL	**4732.24**	**TOTAL**	**535.29**	**TOTAL**	**2614.26**

JULY		AUGUST		SEPTEMBER	
NAME OF THE SCRIPT	**AMOUNT**	**NAME OF THE SCRIPT**	**AMOUNT**	**NAME OF THE SCRIPT**	**AMOUNT**
	NIL	GMR INFRASTRUCTURE LIMITED	3871.39	HOV SERVICES LIMITED	129.68
		TECH MAHINDRA LTD.	1719.73	ATLANTA LIMITED	159.78
				ACTION CONSTRUCTION EQP LTD	147.75
				VOLTAMP TRANSFORMERS LIMITED	458.57
		TOTAL	5591.12	TOTAL	895.78
OCTOBER		**NOVEMBER**		**DECEMBER**	
NAME OF THE SCRIPT	**AMOUNT**	**NAME OF THE SCRIPT**	**AMOUNT**	**NAME OF THE SCRIPT**	**AMOUNT**
GLOBAL VECTRA HELICORP LTD	146.52	PARSVNATH DEVELOPERS LTD.	4696.43	ESS DEE ALUMINIUM LIMITED	456.65
DEVELPMENT CREDIT BANK	492.54	LANCO INFRATECH LTD.	3386.97	XL TELECOM LTD.	165.69
ACCEL FRONTLINE LIMITED	6.14	INFO EDGE (INDIA) LIMITED	679.56	L.T.OVERSEAS LTD.	62.12
HANUNG TOYS AND TEXTILES LTD.	227.92			SOBHA DEVELOPERS LTD.	2030.88
JHS SVENDGAARD LABS LTD	96.84			BLUE BIRD INDIA LIMITED	275.28

FIEM INDUSTRIES LIMITED	134.65			NISSAN COPPER LIMITED	112.50
GWALIOR CHEMICAL INDUS LTD	193.25			RUCHIRA PAPER MILLS LTD	0.00
GAYATRI PROJECTS LIMITED	149.10				
TOTAL	**1446.96**	**TOTAL**	**8762.96**	**TOTAL**	**3103.12**

FII INFLOWS INTO INDIVIDUAL IPO

The process of calculating FII inflows for each script has been explained with the following table. For the month of August 2006 we observe that there were two IPOs, GMR Infrastructure Limited and tech Mahindra Ltd. We obtained the total transaction value of FIIs investing in the IPOs from the SEBI Website on a daily basis. The total sum of daily FII transactions during the period of this IPO was the total FII investment in such IPOs.

Table 14 FII inflows into an individual IPO OF 2006

(EXAMPLE)

SCRIP_NAME	ISIN	TR_DATE	TR_TYPE(*)	RATE	QUANTITY	VALUE (in Rs)
GMR INFRASTRUCTURE LIMITED	INE776C01013	8/18/2006	2	210	340	71400

GMR INFRASTRUCTURE LIMITED	INE776C01013	8/18/2006	2	210	27625	5801250
GMR INFRASTRUCTURE LIMITED	INE776C01013	8/18/2006	2	210	26494	5563740
GMR INFRASTRUCTURE LIMITED	INE776C01013	8/18/2006	2	210	11766	2470860
GMR INFRASTRUCTURE LIMITED	INE776C01013	8/18/2006	2	210	954455	2E+08
GMR INFRASTRUCTURE LIMITED	INE776C01013	8/18/2006	2	210	915375	1.92E+08
GMR INFRASTRUCTURE LTD.	INE776C01013	8/18/2006	2	210	484252	1.02E+08
GMR INFRASTRUCTURE LTD	INE776C01013	8/8/2006	2	210	399618	83919780
GMR INFRASTRUCTURE LTD	INE776C01013	8/14/2006	2	210	100798	21167580
GMR INFRASTRUCTURE LTD	INE776C01013	8/9/2006	2	210	3482663	7.31E+08
GMR INFRASTRUCTURE LTD	INE776C01013	8/11/2006	2	210	42826	8993460
GMR INFRASTRUCTURE LTD	INE776C01013	8/11/2006	2	210	166952	35059920
GMR INFRASTRUCTU	INE776C01013	8/18/2006	2	210	4008196	8.42E+08

RE LTD						
GMR INFRASTRUCTURE LTD	INE776C01013	8/10/2006	2	210	521614	1.1E+08
GMR INFRASTRUCTURE LTD	INE776C01013	8/11/2006	2	210	15539	3263190
GMR INFRASTRUCTURE LTD	INE776C01013	8/8/2006	2	210	536880	1.13E+08
GMR INFRASTRUCTURE LTD	INE776C01013	8/11/2006	2	210	6241	1310610
GMR INFRASTRUCTURE LTD	INE776C01013	8/8/2006	2	210	215604	45276840
GMR INFRASTRUCTURE LTD	INE776C01013	8/16/2006	2	210	8964	1882440
GMR INFRASTRUCTURE LTD	INE776C01013	8/16/2006	2	210	11566	2428860
GMR INFRASTRUCTURE LTD	INE776C01013	8/9/2006	2	210	309725	65042250
GMR INFRASTRUCTURE LTD	INE776C01013	8/11/2006	2	210	9535	2002350
GMR INFRASTRUCTURE LTD	INE776C01013	8/9/2006	2	210	329441	69182610
GMR INFRASTRUCTURE LTD	INE776C01013	8/16/2006	2	210	15097	3170370
GMR	INE776C010	8/10/200	2	210	12845	2697450

INFRASTRUCTURE LTD	13	6				
GMR INFRASTRUCTURE LTD	INE776C01013	8/18/2006	2	210	882431	1.85E+08
GMR INFRASTRUCTURE LTD	INE776C01013	8/18/2006	2	210	7356	1544760
GMR INFRASTRUCTURE LTD	INE776C01013	8/18/2006	2	210	16291	3421110
GMR INFRASTRUCTURE LTD	INE776C01013	8/18/2006	2	210	562874	1.18E+08
GMR INFRASTRUCTURE LTD	INE776C01013	8/18/2006	2	210	25540	5363400
GMR INFRASTRUCTURE LTD	INE776C01013	8/18/2006	2	210	254140	53369400
GMR INFRASTRUCTURE LTD	INE776C01013	8/18/2006	2	210	48240	10130400
GMR INFRASTRUCTURE LTD	INE776C01013	8/18/2006	2	210	74565	15658650
GMR INFRASTRUCTURE LTD	INE776C01013	8/18/2006	2	210	44142	9269820
GMR INFRASTRUCTURE LTD	INE776C01013	8/18/2006	2	210	3577	751170
GMR INFRASTRUCTURE LTD	INE776C01013	8/18/2006	2	210	416835	87535350

GMR INFRASTRUCTURE LTD	INE776C01013	8/18/2006	2	210	20253	4253130
GMR INFRASTRUCTURE LTD	INE776C01013	8/18/2006	2	210	82320	17287200
GMR INFRASTRUCTURE LTD	INE776C01013	8/18/2006	2	210	54781	11504010
GMR INFRASTRUCTURE LTD	INE776C01013	8/18/2006	2	210	1131938	2.38E+08
GMR INFRASTRUCTURE LTD	INE776C01013	8/18/2006	2	210	521729	1.1E+08
GMR INFRASTRUCTURE LTD	INE776C01013	8/18/2006	2	210	164067	34454070
GMR INFRASTRUCTURE LTD	INE776C01013	8/18/2006	2	210	1278	268380
GMR INFRASTRUCTURE LTD	INE776C01013	8/18/2006	2	210	103	21630
GMR INFRASTRUCTURE LTD	INE776C01013	8/18/2006	2	210	1091693	2.29E+08
GMR INFRASTRUCTURE LTD	INE776C01013	8/18/2006	2	210	121063	25423230
GMR INFRASTRUCTURE LTD	INE776C01013	8/18/2006	2	210	12106	2542260
GMR INFRASTRUCTU	INE776C01013	8/18/2006	2	210	248188	52119480

RE LTD						
GMR INFRASTRUCTURE LTD	INE776C01013	8/18/2006	2	210	2159	453390
GMR INFRASTRUCTURE LTD	INE776C01013	8/18/2006	2	210	12064	2533440
GMR INFRASTRUCTURE LTD	INE776C01013	8/18/2006	2	210	586	123060
GMR INFRASTRUCTURE LTD	INE776C01013	8/18/2006	2	210	2383	500430
GMR INFRASTRUCTURE LTD	INE776C01013	8/18/2006	2	210	1396	293160
GMR INFRASTRUCTURE LTD	INE776C01013	8/18/2006	2	210	15100	3171000
GMR INFRASTRUCTURE LTD	INE776C01013	8/18/2006	2	210	1586	333060
					TOTAL	38700,00,000.00

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Thus firstly we calculated the FII inflows into each script. Then we calculated the total of such FII inflows for the entire month relevant to the particular FIIs of that month. Thus we obtained FII inflows for a period of 72 months from 2006 to 2011.But as GDP is available on a quarterly basis we again had to reduce it to quarterly values.

Table 15 Quarterly FII inflows from 2006-2011.(RS MN)

MARCH'06	**13631.28**	MARCH'08	**64309.27**	MARCH'10	**23823.07**
JUNE'06	**7881.79**	JUNE'08	**2312.33**	JUNE'10	**7366.97**
SEP'06	**6486.91**	SEP'08	**7200.53**	SEP'10	**11387.83**
DECEMBER'06	**13313.03**	DECEMBER'08	**129.00**	DECEMBER'11	**80699.12**

MARCH'07	**59184.43**	MARCH'09	**118.98**	MARCH'11	**3232.11**
JUNE'07	**7176.71**	JUNE'09	**.00**	JUNE'11	**3472.28**
SEP'07	**84487.23**	SEP'09	**39572.73**	SEP'11	**4271.42**
DECEMBER'07	**28579.07**	DECEMBER'09	**13982.94**	DECEMBER'11	**1154.06**

. **4.3 DATA ANALYSIS AND INTERPRETATION**

MODEL 1

$NIFTY_i = B + b_1 FII_i + b_2 \ GDP_i + e_i$ …………………………………. (1)

Where,

FII_i is the summation of the Foreign Institutional investments flowing into each respective IPOs for that quarter.

$NIFTY_i$ is the closing monthly values for that quarter.

GDP_i is the quarterly GDP reported.

e_i is the error term are independent normally distributed random variables with zero mean and constant variance σ^2, i=1..n.

b_1 and b_2 are the unknown model parameters.

i=1….n

n=24

RESULTS EXPLAINING MODEL

Table 16: Regression results of FII inflows into IPOs & GDP on Nifty

DEPENDENT VARIABLE – NIFTY

Independent Variables	FII INFLOWS IN IPO	GDP	
Standardized beta coefficient	.325	.668	
Constant			
t ratio	2.121	4.360	
Significance	.046	.000	
Collinearity Tolerance VIF	 .992 1.008	 .992 1.008	
R Square			.512
Adj R Square			.465
Durbin Watson			1.667
F Statistic			11.006
Significance of F statistic			.000

Fig 16 Histogram of regression standardised residual of FII inflows into IPOs & GDP on Nifty

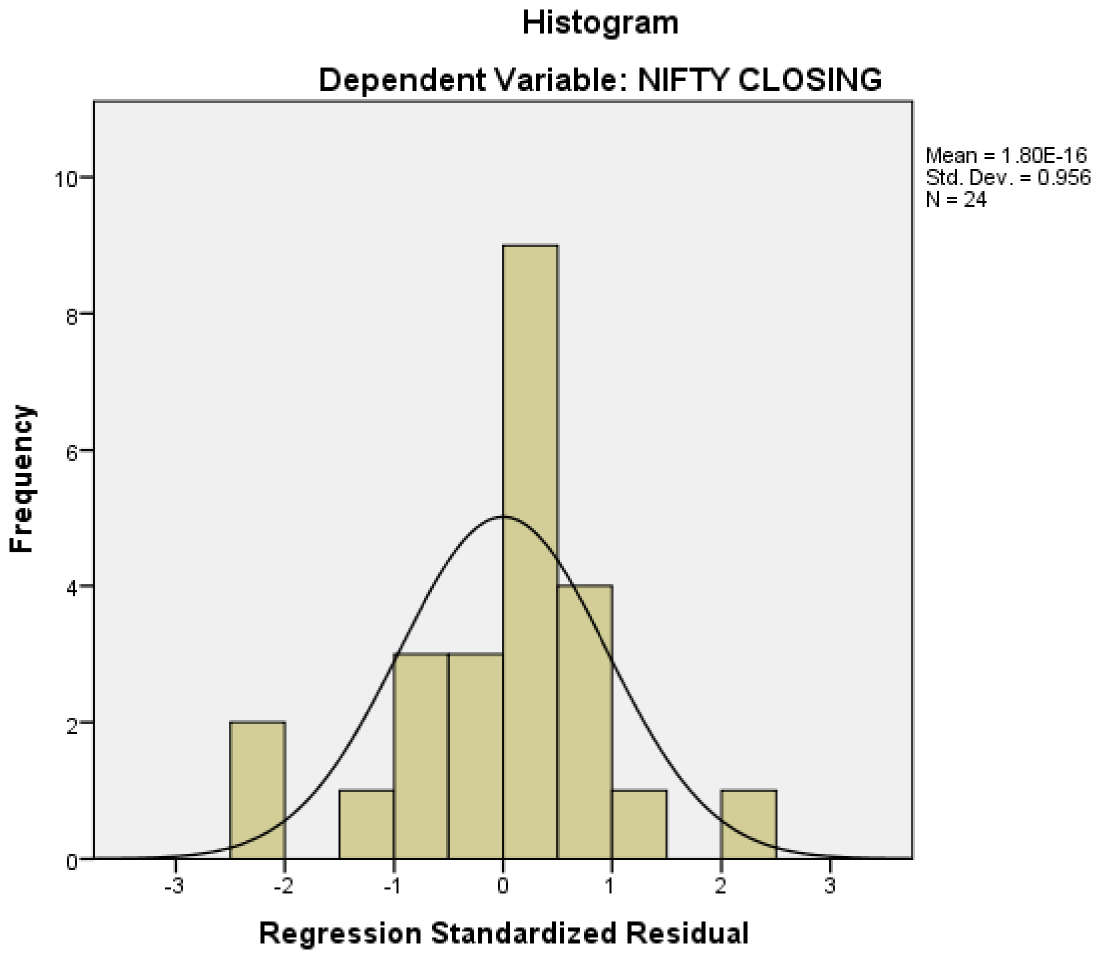
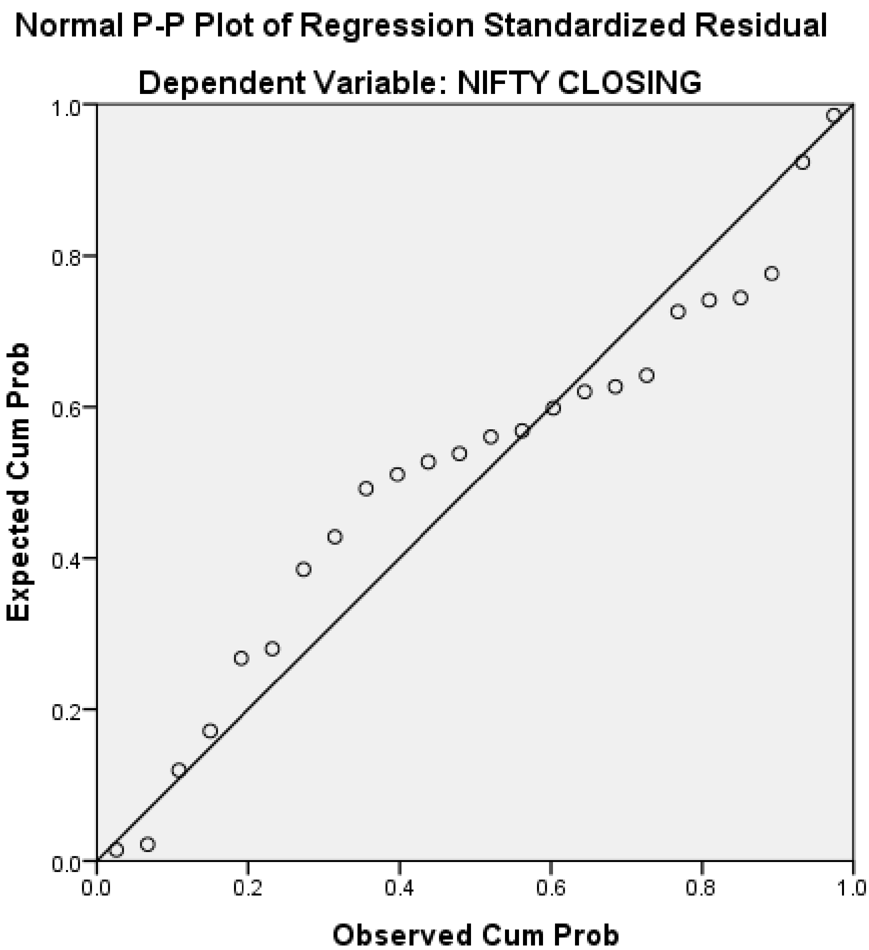


Fig 17 Normal P-P Plot of regression standardised residual of FII inflows Into IPOs & GDP on Nifty.



INTER PRETATION OF THE SUMMARY STATISTICS

From Table 3 we see that R^2 is .512 and the adjusted R^2 is .465.This means that the explanatory variables together account for explaining 51 % of the variation in the response variable Nifty. The difference for the final model is very small i.e .512-.465 = .047 .This shrinkage means that if the model were derived from the population rather than a sample it would account for approximately .047 less variance in the outcome.

The F ratio represents the ratio of the improvement in prediction that results from fitting the model relative to the inaccuracy that still exists in the model. In this case the F ratio is 11.006 which is moderately large and is significant at 1 % level.

The Durbin Watson test statistic to check for serial correlation of the random errors is 1.667 and being closer to 2 implies that the error terms are independent.

We also note from the VIF and Tolerance statistic that multicollineariy is not a serious problem in our model as the tolerance is far above .1 and VIF is far below 10.

Moreover the P-P plot of the regression standardized residual imply that the data set was normally distributed.

INTERPRETATION OF THE REGRESSION PARAMETERS

The b- values tells us about the relationship between Nifty and each of the independent predictors FII investments in IPOs and GDP. The t statistic being significant at 5 % for FII inflows into IPO and at 1 % for GDP implies that the b values are significantly different from zero. In other words it implies that the predictors are making significant contribution to the model.

Thus we reject our null Hypothesis and accept the alternate hypothesis at 5% level significance and arrive at the conclusion that FII inflows into IPOs affect Nifty positively.

MODEL 2

MARKET CAPITALISATION$_i$= B + b_1FII$_i$ +b_2 GDP$_i$ +e_i .... (1)

Where,

FII$_i$ is the summation of the Foreign Institutional investments flowing into each respective IPOs for that quarter.

MARKET CAPITALISATION$_i$ is theclosing monthly values for that quarter.

GDP$_i$ is the quarterly GDP reported.

e_i is the error term are independent normally distributed random variables with zero mean and constant variance σ^2, i=1..n.

b_1 and b_2 are the unknown model parameters.

i=1….n

n=24

Table 17 Regression results of FII inflows into IPOs & GDP on Market Capitalisation

DEPENDENT VARIABLE – MARKETCAPITALISATION

REGRESSION RESULTS

Independent Variables	FII INFLOWS IN IPO	GDP	
Standardized beta coefficient	.272	.794	
Constant			
T ratio	2.142	6.265	
significance	.044	.000	
Collinearity Tolerance VIF	 .992 1.008	 .992 1.008	
R Square			.665
Adj R Square			.633
Durbin Watson			1.527
F Statistic			20.869
Significance of F statistic			.000

Fig 18 Histogram of regression standardised residual of FII inflows into IPOs& GDP on Market capitalisation.

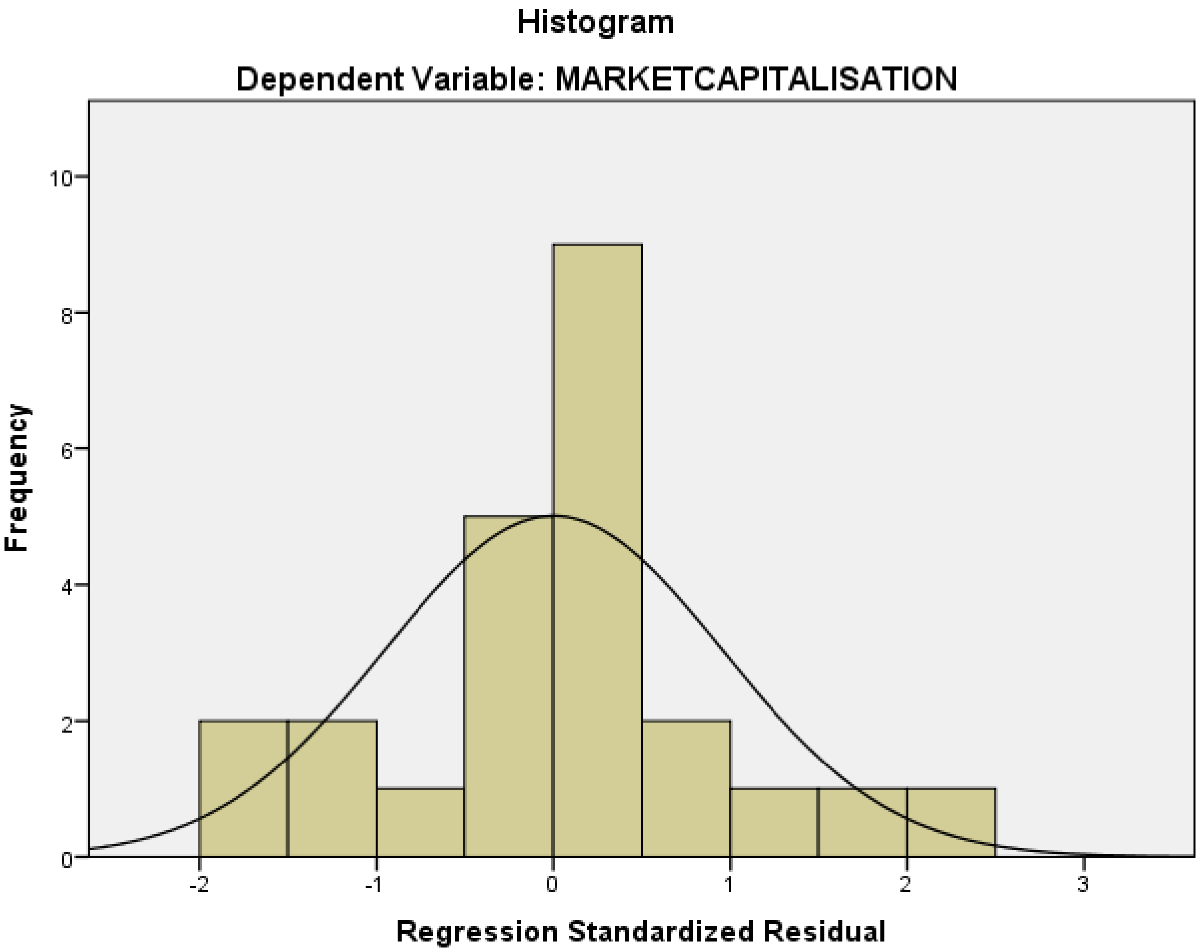


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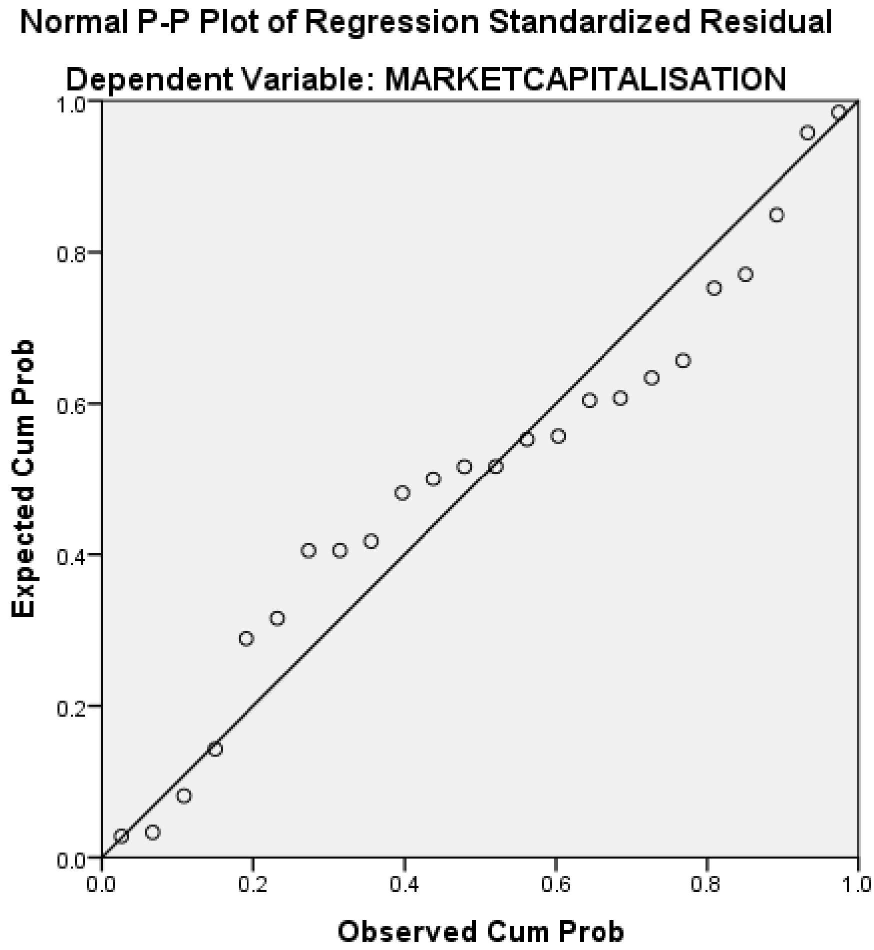
Fig 19 Normal P-P Plot of regression standardised residual of FII inflows Into IPOs & GDP on Market Capitalisation



SECTION 5

FII INVESTMENTS AND SHORT RUN LISTING PERFORMANCE

5.1 INTRODUCTION

It is a common feature all across Globe that the listing performance of the IPOs or the price at which IPOs get listed on the very first day might lead to huge listing gains or even losses. The phenomenon is known as underpricing or overpricing respectively. The degree of underpricing (overpricing) may be measured by the extent to which the first observed price (P) in the secondary market for the stock is above (below) the offer price (OP) of the stock. For example, [(P - OP)/OP] x 100 > 0 indicates underpricing, and [(P - OP)/OP] x 100 < 0 indicates overpricing (Lim et.al, 1990). Underpricing denotes the positive spread between the opening price on the first trading day and the offer price, scaled by the offer price. The resulting number indicates that how many times the respective shares are underpriced. If the number is equal to zero, the firm's shares are called fully priced. If the number has negative sign the firm's shares are called overpriced. The company going for IPOs normally fixes the issue price on the basis of their historical financial performance and market conditions. The share is listed in a recognized stock exchange at a particular price and after listing, the price may move in any directiondepending on the perception of the market. Various studies have been conducted on such such IPO performance behavior.

5.2 RESEARCH HYPOTHESIS AND METHODOLOGY

Hypothesis

Thus based on the above objectives we frame the following research hypothesis:

RH $_{=}$ FII investment proportions have no impact on listing gain.

The study expects to support the view that FII investment proportions positively affect listing gain.

Research Methodology

The time period used in the previous section remains the same for this section as well.

Data

In this section we have used the following data of the following variables for our analysis:

a. FII investment proportion (total FII inflows into each IPO/Volume of IPO)
b. Subscription times (no of times the issue has been subscribed by retail investors)
c. Listing gain or loss.

Table 18 FII investment proportion in each stock ,market subscription& short run Listing gain /loss .

COMPANY NAME	PROPORTION OF FII INVESTMENT	SUBSCRIPTION TIMES	LISTING GAIN/LOSS
BARTRONICS INDIA LTD.	0.219123333	29.08	35.75
CELEBRITY FASHIONS LIMITED	0.247785714	23.6	48.9
EDUCOMP SOLUTIONS LTD.	0.1895	4.18	159.05
PUNJ LOYD LTD	0.432400768	39.06	358.15
TULIP IT SERVICES LTD.	0.194246444	25.08	63.8
PVR LTD DEMAT	0.221131757	1.04	69.95
GVK POWER & INFRASTRUCTURE L	0.258640507	25.82	5.85
INOX LEISURE LIMITED	0.265010364	49.06	55.25
JAGRAN PRAKASHAN LIMITED	0.205408618	14.22	-49.1
GUJARAT STATE PETRONET LTD	0.148457058	48.75	13.45
ENTERTAINMENT NETWORK INDIA LTD	0.284779545	1.26	102.7

ROYAL ORCHIDS HOTELS LIMITED	0.271706012	39	66.4
NITIN SPINNERS LTD.	0.193275002	1.05	5
VISA STEEL LTD	0.435308333	6.6	-0.2
NITCO TILES LTD	0.2747307	4.68	15.25
J.K CEMENT LIMITED	0.4023493	1.76	7.5
MAHINDRA AND MAHINDRA FINANCIAL SERVICES	0.47842615	27	32.55
B.L. KASHYAP AND SONS LTD	0.335468653	15.16	288.3
PRATIBHA INDUSTRIES LTD.	0.215740957	21.34	61.3
GITANJALI GEMS LTD.	0.326625588	15.35	-27.4
SADBHAV ENGINEERING LTD.	0.246638621	19.07	135.65
INDO TECH TRANSFORMERS LTDDM	0.08073807	39.21	109.1
SUNIL HITECH ENGINEERS LTD.	0.057642136	13.14	57.2
SUN TV LTD DM	0.428021548	46.92	589.65
EMKAY SHARE AND STOCK BROKERS LTD.	0.31465472	7.73	16.75
GODAWARI POWER AND ISPAT LTD	0.135415756	4	22.15
R SYSTEMS INTERNATIONAL LIMITED	0.324247786	3.62	-0.35
KEWAL KIRAN CLOTHING LTD.	0.350737419	12.4174	-16.55
UTTAM SUGAR MILLS LTD.	0.43235	4.6329	76.3
ADHUNIK METALIKS LTD.	0.303371991	3.38	4.95
SOLAR EXPLOSIVES LTD.	0.209801364	13.89	74.6
GALLANTT METAL LTD.	0.086107529	2.397	4.35
MALU PAPER MILLS LTD	0	1.54	2.45
SHIVALIK GLOBAL, SPL GLOBAL LTD	0	28	22.55
ROHIT FERRO-TECH LIMITED.	0.459925862	1.72	0.4
PLETHICO PHARMACEUTICALS	0.364658482	25.3142	119.15

LTD.			
LOKESH MACHINES LTD.	0.251224667	19.37	87.4
KAMDHENU ISPAT LTD	0	5.2	20.4
ALLCARGO GLOBAL LOGISTICS PRIVATE LTD	0.407228002	7.64	-5.4
PRIME FOCUS LTD.	0.354869919	1.34	-90.85
DECCAN AVIATION LIMITED	0.228275741	1.13	-48.9
UNITY INFRAPROJECTS LTD.	0.369462678	2.3	-204.1
GMR INFRASTRUCTURE LIMITED	0.483394202	6.6.8	0.4
TECH MAHINDRA LTD.	0.369653146	72	189.25
HOV SERVICES LIMITED	0.160096543	2.51	-19.7
ATLANTA LIMITED	0.247719767	9.45	42.3
ACTION CONSTRUCTION EQP LTD	0.049415957	33	63.5
VOLTAMP TRANSFORMERS LIMITED	0.272162274	17.4	71.45
GLOBAL VECTRA HELICORP LTD	0.226281429	3.53	2.5
DEVELPMENT CREDIT BANK	0.264951203	35.26	21.45
ACCEL FRONTLINE LIMITED	0.01453654	2.19	-4.5
HANUNG TOYS AND TEXTILES LTD.	0.252538737	8.83	1.7
JHS SVENDGAARD LABS LTD	0.269287097	2.61	-1.05
FIEM INDUSTRIES LIMITED	0.239724634	2.539	-15.45
GWALIOR CHEMICAL INDUS LTD	0.24155901	7.2983	12.85
GAYATRI PROJECTS LIMITED	0.364631976	5.68	157.85
PARSVNATH DEVELOPERS LTD.	0.470989831	56.2661	226.4
LANCO INFRATECH LTD.	0.317328814	1.8	1.4
INFO EDGE (INDIA) LIMITED	0.398890369	54.786	272.55
ESS DEE ALUMINIUM LIMITED	0.291600718	31.63	13.3

XL TELECOM LTD.	0.279170483	8.036	-14.65
L.T.OVERSEAS LTD.	0.157658925	6.94	-2.8
SOBHA DEVELOPERS LTD.	0.356812203	113.92	286.65
BLUE BIRD INDIA LIMITED	0.29877151	4.82	-10.35
NISSAN COPPER LIMITED	0.44999994	5.25	91.9
RUCHIRA PAPER MILLS LTD	0	2.534	-1.95
AUTOLINE INDUSTRIES LTD	0.242061867	17.36	32.15
SHREE ASHTAVINAYAK CINE VISION LIMITED	0.27340303	6.04	65.4
Pyramid Saimira Theatre Limited	0.344326306	16.26	57.7
CAIRN INDIA LTD.	0.574443162	1.14	-22.6
TANLA SOLUTIONS	0.34484734	38.65	114.8
RAJ RAYON		3.8602	
LUMAX AUTO TECHNOLOGIES LIMITED	0.105525564	3.23	35.7
C AND C CONSTRUCITONS LTD	0.412108737	20.25	-49.55
SMS PHARMACEUTICALS LIMITED	0.335789678	2.64	-20.75
POWER FINANCE CORP	0.33500748	77.24	26.65
TRANSWARRANTY FINANCE LTD	0.297022833	1.59	-4.55
FIRSTSOURCE SOLUTION LTD.(EX-ICICI ONE S	0.425840462	50.07	15.55
REDINGTON INDIA LIMITED	0.383445847	43.27	50.7
CINEMAX INDIA LTD	0.260245	42.32	-2.7
TECHNOCRAFT INDS(INDIA)LTDDM	0.192149038	10.67	-4.95
HOUSE OF PEARLS FASHIONS LTD	0.482610155	3.91	-80.05
AKRUTI NIRMAN LTD	0.428683134	81.05	23.45
GLOBAL BROADCAST NEWS LIMITED	0	48.74	255.9
POCHIRAJU INDUSTRIES LTD.	0.459926555	3.77	19.65

CAMBRIDGE TECHNOLOGY ENTERPRISES LIMITED	0.143995325	6.28	71.9
ABHISHEK MILLS LTD DM	0.429872439	1.26	-8.75
PAGE INDUSTRIES LTD.	0.50132525	1.44	-88.2
RAJ TELEVISION NETWORK LIMITED	0.03362994	3.04	-31
AMD METPLAST LTD DM	0.438087862	5.32	2.9
IDEA CELLULAR (EX BIRLA AT&T)	0.437690658	49.51	10.7
EVENIX ACCESSOIRES	0	3.7	-45.95
MUDRA LIFESTYLE LIMITED	0.266228288	4.59	-26.05
ORIENTAL TRIMEX LIMITED	0.122087766	1.05	-18.3
MINDTREE CONSULTING LTD.	0.317486457	103.28	196.2
BROADCAST INITIATIVES LTD DM	0.410919181	2.76	-49.6
INDUS FILA LTD	0.169979355	1.43	-36.15
EURO CERAMICS LTD DM	0.324113445	3.02	-45.85
INDIAN BANK	0.341564259	32.16	7.35
ASTRAL POLYTECNIC LTD	0	1.7	-9.65
ADVANTA INDIA LIMITED	1.027611446	3.98	205.5
ICRA LIMITED	0.21881058	75.04	473.25
ORBIT CORPORATION LTD.	0.351711229	3.85	18.2
INSECTICIDES INDIA LTD	0	1.97	-5.55
BINANI CEMENTS LTD	0.493959021	1.36	-5.95
MIC ELECTRONICS	0.285664902	50.59	188.15
BHAGWATI BANQUETS & HOTELSDM	0.211996218	1.83	9.05
FORTIS HEALTHCARE LTD	0.340922316	2.78	-7.85
HILTON METAL FORGING LIMITED	0.215562421	1.06	-2.25
NELCAST LIMITED-DO NOT SELL	0.317531494	7.36	-11.65
MEGHMANI ORGANICS LTD	0.420801587	23.94	7.6

DECOLIGHT CERAMICS LIMITED	0.155478776	1.53	-9.35
TIME TECHNOPLAST LTD	0.366444194	49.55	165.9
NITIN FIRE PROTECTION INDUSTRIES LTD	0.225148378	48.48	294.85
ASAHI SONGWON COLORS LTD DM	0.476732687	1.85	-0.05
GLORY PLYFILMS LTD	0	1.08	14.1
ALLIED DIGITAL SERV LTD DM	0.219171798	60.87	140.15
BHARAT EARTH MOVERS LIMITED	0.21	30.65	121
HOUSING DEVELOPMENT N INFRASTRUCTURE LTD	0.43582942	30.65	59.35
SURYACHAKRA POWER CORPORATION LIMITED	0.966058298	6.6	2.35
ANKIT METAL AND POWER LTD.	1.030820571	2.18	0.95
ICICI BANK LTD	0.09219917	1.58	41.55
ROMAN TARMAT LTD DM	0.344821034	29.67	144.2
DLF LIMITED	0.576820274	3.47	44.8
VISHAL RETAIL	0	69.08	483.1
CELESTIAL LABS LTD	0.64753	2.67	7.55
SEL MFG CO LTD DM	0.282397745	3.62	54.75
ASIAN GRANITO INDIA LTD.	0.156427714	4.51	-2.8
TAKE SOLUTION LTD.	0.352119524	59.48	197.8
K.P.R.MILL LTD.	0.208088289	1.19	-50.8
PURVANKARA PROJECTS LIMITED	0.545671215	1.91	-37.7
CENTRAL BANK OF INDIA	0.3417692	62	13.3
PRIME URBAN DEVELOPERS	0	5.75	-92.1
OMNITECH INFOSOLUTIONS LTD.	0.2264226	61.84	58.4
ZYLOG SYSTEMS LTD.	0.279826667	76.51	77.5

OMAXE LTD.	0.372944691	68.34	39.35
ALPA LABORATORIES LTD.	0.087094737	1.11	-12.85
SIMPLEX LTD	0.3	85.53	88.7
EVERONN SYSTEMS INDIA LTD.	0.21570444	131.47	337.35
MOTILAL OSWAL FINANCIAL SERVICES LIMITED	0.405537134	27.41	151.85
INDOWIND ENERGY LTD.	0.390048	0.97	48.65
MAGNUM VENTURES LTD	0.421146989	2.95	19.4
MAYTAS INFRA LIMITED.	0.382514576	67.86	243.35
SUPREME INFRASTUCTURE INDIA LTD.	0.242828777	52.75	67.1
KOUTONS RETAIL INDIA LTD	0.34123446	45.52	171.5
CONSOLIDATED CONSTRUCTION CONSORTIUM LTD	0.314947297	81.18	282.1
KAVERI SEED COMPANY LIMITED	0.19529425	4	60.95
DHANUS TECHNOLOGIES LIMITED	0.49556558	28.47	24.75
POWER GRID	0.330609821	64.82	48.6
MUNDRA P AND E Z LTD RS10 DM	0.400121913	115.84	522.9
EMPEE DISTILLERIES LIMITED	0.414331458	6.87	-80.65
RELIGARE ENTERPRISES LTD	0.34963336	160.56	340.3
BARAK VALLEY CEMENTS LIMITED	0.233213074	29.15	13.3
VARUN INDUSTRIES LTD.	0.037	36.46	52.2
BRIGADE ENTERPRISES LIMITED	0.321476059	13.07	-10.1
TRANSFORMERS AND RECTIFIERS (INDIA) LTD	0.240354452	91.31	264.25
ECLERX SERVICES LIMITED	0.330117505	26.3	134.65

JYOTHY LABS LTD.	0.209714451	45.83	104.05
KAUSHALYA INFRA DEV CORP LDM	0.007051455	7.2	22.45
KOLTE PATIL DEVELOPERS LIMITED.	0.270950684	46.18	36.35
RENAISSANCE JEWELLERY LIMITED WARRANT	0.263079063	23.63	14.9
EDELWEISS CAPITAL LTD	0.360413908	110.96	685.25
PRECISION PIPES	0	10.35	-13.9
MANAKASI LTD	0	8.98	7.7
ARIES LTD	0	7.62	121.4
BGR ENERGY SYSTEMS LTD	0.317300898	119.54	421.45
BURNPUR CEMENT LTD	0.012690366	14.79	36.05
TULSI EXTRUSION LIMITED	0.488375263	2	55.85
IRB INFRASTRUCTURE DEVELOPERS LIMITED	0.503953078	4.3	4.65
SHRIRAM EPC LIMITED	0.2694726	3.91	-13.5
BANG OVERSEAS LIMITED	0.485714286	1.24	-32.9
ONMOBILE GLOBAL LTD DM	0.441540183	10.95	78.15
KNR CONSTRUCTION LTD DM	0.213523038	1.25	-15.1
CORDS CABLE INDUSTRIES LIMITED	0.294003548	4.99	4.45
J.KUMAR INFRA	0	2.17	-6.65
RELIANCE POWER LIMITED	0.45809414	73.04	-77.7
FUTURE CAPITAL HLDS LTD DM	0.37969875	133.44	144.8
RURAL ELECTRIFICATION CORPORATION LIMITE	0.306772915	27.76	16.3
V-GUARD INDUSTRIES LIMITED	0.07843775	2.7	-6.05
GSS AMERICA INFOTECH LIMITED.	0.500426079	1.08	100.8
KIRI DYES AND CHEMICALS	0.037225899	1.43	8.95

LIMITED			
TITAGARH WAGONS LTD.	0.321129744	6.75	166.85
SITA SHREE FOOD PRODUCTS LTD	0.443981333	2.44	13.7
GAMMON INFRASRUCTURE PROJECTS LTD.	0.414325861	3.48	-8.85
GOKUL REFOILS & SOLVENT LTDM	0.289638331	4.27	-12.95
BIRLA COTSYN (INDIA) LIMITED	0.005397015	1.11	-4.55
KSK ENE VEN LTD DM	0.606244633	1.5	-48.25
LOTUS EYE CARE HOSPITAL LIMITED	0.26355	1.18	-2.35
FIRST WINNER INDUSTRIES LTD	0.490901951	50.07	-35.8
ARCHIDPLY INDUSTRIES LTD.	0.100001123	1.52	-23.3
SEJAL ARCHITECTURAL GLASS LTD	0.347674846	9.9	-33.75
VISHAL INFORMATION TECHNOLOGIES LIMITED	0.200659309	1.19	44.6
NU TEK INDIA LIMITED	0.149032889	1.63	7.15
RESURGERE MINES AND MINERALS INDIA LTD	0.538213258	1.16	263.55
AUSTRAL COKE & PROJECTS LIMITED	0.311502755	1.65	29.95
20 MICRONS LIMITED	0.146233077	4.29	-21.35
ALKALI METALS LIMITED	0.357907843	1.04	70.4
EDSERV SOFTSYSTEMS LTD	0.499027975	1.3	77.7
MAHINDRA HOLI N RESORTS IND	0.018257758	9.8	17.45
ADANI POWER LTD	0.324590777	21.64	-88.9
RAJ OIL MILLS LTD	0	1.24	-0.75
EXCEL INFOWAYS LTD	0	1.97	10.85

OIL INDIA LTD	0.348406867	30.82	91.2
GLOBUS SPIRITS LTD	0.106661333	1.5	-9
JINDAL COTEX LTD	0.239921651	2.2	12.3
NHPC LTD	0.327071527	23.74	0.75
INDIABULLS POWER LTD	0.589543081	21.84	-5.5
THINKSOFT GLOBAL SERVICES LIMITED	0.084393856	2.57	39.4
EURO MULTIVISION LIMITED	0.496492045	1.81	-21.45
PIPAVAV SHIPYARD LTD.	0.283500972	8.25	-1.3
ASTEC LIFESCIENCES LIMITED	0.176217135	1.56	2
DEN NETWORKS LIMITED	0.394315979	1.04	-31.6
COX N KINGS(I)LTD(EX-COX N KINGS(I) PVT	0.26915937	6.31	95.4
MBL INFRASTRUCTURE LTD	0.073684211	1.97	26.55
D.B. CORPORATION LTD	0.114189439	32.5	53.9
GODREJ PROPERTIES LTD	0.368168011	4	47.25
JSW ENERGY LTD DM	0.249915067	1.68	0.85
INFINITE COMPUTER SOLUTIONS (INDIA) LTD	0.298559711	43.22	26.8
JUBILANT FOODWORKS LIMITED	0.294746613	31.11	84.1
SYNCOM HEALTHCARE LIMITED	0.493056	5.17	12.75
VASCON ENGINEERS LTD	0.110899722	1.216	-16.95
THANGAMAYIL JEWELLERY LIMITED	0.260842347	1.117	-3.95
AQUA LOGISTICS LTD	0.1028234	1.93	24.6
D B REALTY LTD DM	0.466622692	2	-11.8
EMMBI POLYARNS LIMITED	0.190467467	0.941	-16.25
HATHWAY CABLE AND DATACOM L	0.209642342	1.36	-32.35
ARSS INFRA PROJ LTD DM	0.332985585	47.62	287.45
TEXMO PIPES AND PRODUCTS	0.4968576	7.48	47.15

LIMITED			
MAN INFRACONSTRUCTION LTD	0.378897213	62.52	97.85
UNITED BANK OF INDIA	0.289325637	33.37	2.65
IL AND FS TRANSPORTATION NETWORK LTD	0.312097514	33	16.65
GOENKA DIAMOND AND JEWELS LIMITED	0	1.07	-7.4
INTRASOFT TECHNOLOGIES LTD	0.33045338	18.95	14.35
SHREE GANESH JW HOUSE LTD DM	0.426381148	1.96	-95.45
PERSISTENT SYSTEMS LIMITED.	0.356477488	93.6	96.35
PRADIP OVERSEAS LIMITED	0.371474245	14.08	-2.95
JAYPEE INFRATECH LTD DM	0.04452539	1.24	-10.55
SJVN LIMITED DM	0.093348382	6.64	-0.9
MANDHANA INDUSTRIES LIMITED	0.318611205	6.32	3.55
TARAPUR TRANSFORMERS LIMITED	0.015668887	1.74	-17.6
NITESH ESTATES LTD DM	0.515163533	1.16	-2.6
TALWALKAR BETTER VALUE FITNESS LTD	0.178340996	28.39	35.15
PARABOLIC DRUGS LTD	0.333694913	1.04	-10.15
TECHNOFAB ENGINEERING LTD	0.25577841	12.78	56.95
HINDUSTAN MEDIA VENTURE LTD	0.131021229	5.43	22.95
Aster Silicates Limited	0	4.47	87.55
PRAKASH STEELAGE LTD	0.401351948	4.53	75.35
BAJAJ CORP LTD	0.217386444	19.29	98.75
SKS MICROFINANCE LTD DM	0.411908822	13.69	103.65

INDOSOLAR LTD	0.321275988	1.55	-5.3
GUJARAT PIPAVAV PORT LIMITED	0.253727969	19.94	8.05
MICROSEC FINANCIAL SERVICES LTD	0.28954608	12.2	-6.65
CAREER POINT INFOSYSTEMS LTD	0.232380852	47.39	318.15
EROS INTERNATIONAL LTD	0.32425905	26.51	15.25
ELECTROSTEEL STEEL LTD	0.24699875	8.23	0.25
ORIENT GREEN POWER COMPANY LTD	0.438998502	1.07	-2.3
RAMKY INFRASTRUCTURE LTD-	0.313423302	2.89	-62.6
GALLANT ISPAT LTD	0.187799136	1.44	31.05
CANTABIL RETAIL INDIA LIMITED	0.163367871	2.35	-30
TECPRO SYSTEMS LTD.	0.263540392	24.47	50.7
VA TECH WABAG LTD	0.226097241	36.22	397.95
ASHOKA BUILDCOM LTD-	0.243659232	15.94	6.75
BEDMUTHA INDUSTRIES LIMITED	0.332865	7.69	77.15
COMMERCIAL ENGINEERS & BODY BUILDERS CO	0.733361132	2.07	-14.1
OBEROI REALTY LTD	0.471843	12.13	22.9
B S TRANSCOMM LIMITED	0.081798319	1.1	133.25
GYSCOAL ALLOYS LIMITED	0.499725777	8.59	10.65
PRESTIGE ESTATE AND PROJECT LTD	0.850321914	2.26	10.15
COAL INDIA LTD	0.311265335	15.28	97.55
GRAVITA INDIA LIMITED	0.417654444	42.88	84.7
RPP INFRA PROJECTS LIMITED	0.046254098	2.97	-6.1
MOIL LTD DM	0.184288515	56.43	90.05
A2Z MAINTAINANCE AND	0.298375934	0.96	-71.45

ENGGINEERING SERVIC			
RAVI KUMAR DISTILLERIES LIMITED	0	2.22	16.05
PUNJAB AND SIND BANK	0.171818539	50.75	7.15
C.MAHENDRA EXPORTS	0.5	2.78	110.25
SHEKHAWATI POLY YARN LIMITED	0.043568417	13.11	47.45643158
OMKAR SPECIALITY CHEMICALS LTD	0.284661612	4.67	-51.55
ACROPETAL TECHNOLOGIES LIMITED	0.499999976	1.28	8.45
SUDAR GARMENTS LIMITED	0.08551818	1.55	36.1
LOVABLE LINGERIE LTD	0.10436038	35.21	44.55
PTC INDIA FINANCIAL SERVICES LIMITED	0.26735281	1.7	-3.1
SHILPI CABLE TECH LTD DM	0.007243422	3.48	-20.95
MUTHOOT FINANCE LTD EQ FV RS 10	0.291461669	24.55	0.9
FUTURE VENTURES INDIA LIMITED	0.0468416	1.52	-1.8
PARAMOUNT PRINTPACKAGING LTD	0.15666321	3.92	-7.95
SERVALAKSHMI PAPER LIMITED	0.116686333	1.47	-9.95
INNOVENTIVE INDUSTRIES LTD	0.089208464	1.24	-22.95
AANJANEYA LIFECARE LIMITED	0.06966	1.11	77.1
SANGHVI LTD	0	1.3	27
TIMBOR HOME LIMITED	0.312323176	5.78	28.55
RUSHIL DECOR LIMITED	0.110846305	2.62	47.5
BHARTYA GLOBAL INFOMEDIA LTD	0	1.47	-52.1

INVENTURE GROWTH & SECURITIES LIMITED	0.1	4.58	89.75
L&T FINANCE HOLDINGS LTD	0.193983517	5.34	-1.95
TREE HOUSE EDUCATION AND ACCESSORIES LTD	0.076160473	1.85	-17.4
TD POWER SYSTEMS LIMITED	0.482958999	2.92	19.25
PG ELECTROPLAST LIMITED	0.491488806	1.34	205.3
BROOKS LABORATORIES LIMITED	0	1.6	-38.5
SRS LIMITED	0	1.25	-24.75
VASWANI INDUSTRIES LIMITED	0	4.16	-30.6
PRAKASH CONSTROWELL LIMITED	0.050025	2.21	92
TIJARIA POLYPIPES LTD	0.2836001	1.2	-41.4
ONELIFE CAPITAL ADVISORS LIMITED	0.5	1.53	35.95
TAKSHEEL SOLUTIONS LTD DM	0.121213636	2.99	-91.85
FLEXITUFF INTERNATIONAL LIMITED	0.195591111	1.17	10.55
M AND B SWITCHGEARS LIMITED	0.5	1.57	132.4
INDO THAI SECURITIES LIMITED	0	1.18	-50.85

ANLYTICAL TOOLS AND TECHNIQUES

In this chapter we attempted to find out whether the two important variables namely FII investment proportion, and the number of times the issue has been subscribed can explain the happening or non-happening of IPO listing gains or losses. We can thus have loss or gain associated to our outcome variable that is listing performance. In such situations we can apply the logistic regression analysis.

The Logistic Regression Model

Regression analysis is used to determine the magnitude of relationships between variables as well as to model relationships between variables and for predictions based on the models. Simple linear regression or multiple linear is applicable when this relationship is assumed to be linear .Logistic regression is multiple regression but with an outcome variable that is a categorical variable and predictor variables that may be categorical or continuous. When we are trying to predict membership of only two categorical outcomes the analysis is known as binary logistic regression but when we want to predict membership of more than two categories we use multinomial logistic regression.

Logistic regression could predict the likelihood, or the odds ratio, of the outcome based on the predictor variables, or covariates. The significance of logistic regression can be evaluated by the log likelihood test, given as the model chi-square test, evaluated at the $p < 0.05$ level, or the Wald statistic. Logistic regression has the advantage of being less affected than discriminant analysis when the normality of the variable cannot be assumed. It has the capacity to analyze a mix of all types of predictors. Logistic regression, which assumes the errors are drawn from a binomial distribution, is formulated to predict and explain a binary categorical variable instead of a metric measure. In logistic regression, the dependent variable is a *log odd* or *logit*, which is the natural log of the odds. Logistic regression allows one to predict a discrete outcome, such as group membership, from a set of variables that may be continuous, discrete, dichotomous, or a mix of any of these. Generally, the dependent or response variable is dichotomous, such as presence/absence or success/failure. In instances where the independent variables are categorical, or a mix of continuous and categorical, logistic regression is preferred. Since the probability of an event must lie between 0 and 1, it is unrealistic to model probabilities with linear regression techniques, because the linear regression model allows the dependent variable to take values greater than 1 or less than 0. The logistic regression model is a type of generalized linear model that extends the linear regression model by linking the range of real numbers to the 0-1 range.

Logistic regression analysis does not require the restrictive assumptions regarding normality distribution of independent variables or equal dispersion matrices nor the prior probabilities of failure. Rather, logistic regression is based on two assumptions; (1) it requires the dependent variable to be dichotomous, with the groups being discrete, non-overlapping, and identifiable; and (2) it considers the cost of type I and type II error rates in the selection of the optimal cut-off probability. βs are the regression coefficients that are estimated through an iterative maximum likelihood method.

However, because of the subjectivity of the choice of these misclassification costs in practice, most researchers minimize the total error rate and, hence, implicitly assume equal costs of type I and type II errors.

In logistic regression model instead of predicting the value of a variable Y from a predictor variable X_1 or several predictor variables Xs, we predict the probability of Y occurring given the known values of X_1 or Xs. Thus where there are several predictors,

$$P(Y) = 1/1+e^{-(b_0+b_1x_{1i}+b_2x_{2i}+\ldots+b_nx_{ni})} \ldots\ldots\ldots\ldots\ldots\ldots(1)$$

Where, P (Y) is the probability of Y occurring, e is the base of natural logarithm, b_0 is the constant, b_n is the coefficient for x_n and i is the i^{th} case. The logistic regression above is the multiple linear regression in logarithmic terms (called *logit)* and this overcomes the problem of violating the assumption of linearity.

5.3 **DATA ANALYSIS AND INTERPRETATION**

Application of the model

We get the following output by running the logistic regression in SPSS 21.

Dependent Variable Encoding

Original Value	Internal Value
loss	0
profit	1

The above table shows that all cases with loss during listing has been coded as 0 and all cases with profit during listing has been coded as 1 during listing to have a binary outcome.

BASIC MODEL

Classification Table from the basic model.

Observed			Predicted		
			PERFORMANCE		Percentage Correct
			loss	profit	
Step 0	PERFORMANCE	loss	0	108	0.0
		profit	0	197	100.0
	Overall Percentage				64.6

a. Constant is included in the model.

b. The cut value is .500

Iteration History

Iteration		-2 Log likelihood	Coefficients
			Constant
Step 0	1	396.489	.584
	2	396.468	.601
	3	396.468	.601

The above table shows that with the basic model with only a constant the model correctly classifies 64.6 % of the outcomes. However since the number of scripts making profits were more in number it is better to predict that all scripts will have profits as it results in greater number of correct predictions.

Variables in the equation

	B	S.E.	Wald	df	Sig.	Exp.(B)
Step 0 Constant	.601	.120	25.203	1	.000	1.824

Variables not in the Equation

			Score	df	Sig.
Step 0	Variables	FIIPROPORTION	4.688	1	.030
		SUBSCRIPTIONTIMES	41.103	1	.000
		FIIPROPORTION by SUBSCRIPTIONTIMES	30.493	1	.000
	Overall Statistics		50.058	3	.000

The value of the constant in the basic model is .60.The table labeled as variables not in the equation tells us that the residual chi-square statistic which is 50.058 is significant at $p<.05$.It means that the coefficients not included in the model are significantly different from zero- and the addition of one or more of these variables will significantly affect its predictive power.

In logistic regression there is an important statistic known as the Wald statistic, which has a special distribution known as the chi square distribution. Like t test in linear regression the Wald statistic tells us whether the b coefficient for that predictor is significantly different from zero or not. If the coefficient is significantly different from zero then we assume that the coefficient is significantly making contribution to the outcome(Y).The Score statistic is similar to the Wald statistic and is used at this stage of the analysis as it's easy to compute in case of large samples.

Thus we see from the significance values of the score statistic that all the above variables will make significant contribution to the model.

Model Summary

Step	-2 Log likelihood	Cox & Snell R Square	Nagelkerke R Square
1	311.194[a]	.244	.335

a. Estimation terinated at iteration number 7 because parameter estimates changed by less than .001.

The above model summary was obtained after all variables were considered in it. If we look at the log likelihood statistic

Chi square =-2LL base model – 2ll new model

Df = (parameters new model – parameters old model)

=2.

SPSS calculates the Cox Snell R square and the Nagelkerke R Square to predict how well the model fits the data and is quite similar to R^2 used in multiple linear regression. Thus we see that the predictive power of the model is moderately good.

Classification Table[a]

	Observed		Predicted		
			PERFORMANCE		Percentage Correct
			loss	profit	
Step 1	PERFORMANCE	loss	54	54	50.0
		profit	37	160	81.2
	Overall Percentage				70.2

a. The cut value is .500

The model correctly classifies 54 of the cases that had made losses but misclassifies 54 others. It also correctly classifies 160 of the cases that had made profit but misclassifies 37 others. The overall accuracy of the classification is 70.2% compared to the basic model which was only 64.6 %.

Table 19 LOGISTIC REGRESSION RESULTS OF FII INVESTMENT PROPORTION & MARKET SUBSCRIPTION ON SHORT RUN LISTING GAIN/LOSS.

		B	S.E.	Wald	df	Sig.	Exp.(B)
Step 1[a]	FIIPROPORTION	3.059	.912	11.241	1	.001	21.306
	SUBSCRIPTIONTIMES	.239	.053	20.066	1	.000	1.270
	FIIPROPORTION by SUBSCRIPTIONTIMES	-.475	.132	13.004	1	.000	.622
	Constant	-1.210	.324	13.926	1	.000	.298

a. Variable(s) entered on step 1: FIIPROPORTION, SUBSCRIPTIONTIMES, FIIPROPORTION * SUBSCRIPTIONTIMES.

The above table is crucial because the Wald statistics are significant implying that the coefficients are significantly different from zero. We need to replace the b values in eqn (1) above to establish the probability that a case falls into a certain category. The most important is the odd ratio Exp. (B).

Odds = P (event)/P (no event).

P (no event) = 1- P (event Y).

P (event Y) $=1/1+e^{-(b_0+b_1x_{1i}+b_2x_{2i}+\ldots+b_nx_{ni})}$

Δ Odds = odds after a unit change in the predictor/ original odds.

This is the odds ratio and if its value is greater than 1 then it indicates that as the predictor increases the odds of the outcome occurring increases. Conversely a value less than 1 indicates that as the predictor increases the odds of the outcome occurring decreases. In this case we can say that odds of a script having profit is 21.306 times higher than a script having no FII investments. Similarly the odds of a script having profit is 1.270 times higher than a script with no market subscription. Thus we see that FII investment proportion makes a significant contribution to the model for chances of profit to happen.

CONCLUSION

As we know that the FIIs are major institutions who generally invest in the foreign country for reaping profits for their investors, they are presumed to invest through well researched strategies whether short term or long term. This research is intended in understanding mainly the various ways in which the FII investments in India affect the stock markets. As discussed through review of literature in the previous sections we observe that there has not been much research that we could witness on FII investments in the Indian Primary market. We have mainly focused on FII investments in Indian IPOs and how this is affecting other significant stock market variables.

In the CHAPTER 3 we tried to see the impact of IPO volumes on the market capitalisation of NSE and also on Nifty. We witnessed that the IPOs considerably affect the market. This is majorly because market capitalization is a product of number of shares and price of the entire NSE. With an IPO the number of shares increases and so market capitalization increases.

CHAPTER 4 aims in explaining the impact of FII investments in IPO on the market capitalization of NSE. The NSE market capitalization is a strong indicator of the overall financial growth of an economy. On the other hand GDP is a strong indicator of the overall economic growth. The objective of considering GDP as a regressor for this purpose was to understand the relative macroeconomic importance of FIIs investing in the IPOs of India in shaping the financial growth of the economy. In our analysis we find that the predictive capacity of the GDP is far more dominant as compared to FII investments in IPO. But the beta values of both suggests that the predictive capacity of GDP is approximately three times more as compared to that of FII investments in IPO. This fact is important to note that though GDP stands as an extremely fundamental factor in predicting the financial growth of an economy comparatively the FIIs have also played a significantly important role in dominating the financial sector of the country. However if FII investments in IPO can affect the market capitalisation of NSE it definitely should have a strong influence on the Nifty itself which was also witnessed.

Moreover with the investments of the foreign investors as examined in CHAPTER 5 we also see that scripts containing foreign investments are more prone to listing gain and so there can be an overall thrust of prices upwards. Thus we see that FII investments in IPOs apart from being an important capital market determinant is also an important criteria in the micro level for having caused significant impact on the listing performance of the stocks in a positive way. As has already been pointed out that various firm level and other macro aspects affect the listing performance of the stocks as per literature this add a new dimension to the fact that FII investment proportions can also adequately affect the performance of the stocks. This may act as a guiding factor to the retail investors and other categories of investors before making investments in stocks in the primary market. Thus FII investments in IPO have a vast role to play both in the macro and micro aspects.

List of Figures:

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